

Clifford geometric algebra: Real and complex spinor data tables

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The modern algebra concepts are used to construct tables of algebraic spinors related to Clifford algebra multivectors with real and complex coefficients. The following data computed by *Mathematica* are presented in form of tables for individual Clifford geometric algebras: 1. Initial idempotent; 2. Two-sided ideal; 3. Left ideal basis (otherwise projector, or spinor basis); 4. Matrix representations (reps) for basis vectors in Clifford algebras in spinor basis; 5. General spinor; 6. Spinor in matrix form; 7. Squared hermitian norm of the spinor. Earlier in 1998, only the first four items computed by *Maple* were published by R. Abłamowicz².

I. TERMS AND DEFINITIONS

Below, the definitions of algebraic concepts on which the computation is based are reminded. The starting structures are the ring $\mathcal{R}$ and ideal $\mathcal{I}$ the properties of which are summarized below. The small circle "o" denotes a non-commutative multiplication. In geometric algebra (GA) $Cl_{p,q}$, where (p,q) and $n = p + q$ define signature and dimension of the vector space, respectively, the circle represents the geometric product (GP) that usually is omitted in formulas.

Division ring $\mathcal{R}$ consists of a set of elements $A, B, C \dots \in \mathcal{R}$ and binary operations "+" (summation) and "o" (multiplication) defined on $\mathcal{R}$. The properties of $\mathcal{R}$ are summarized in Table I. The division ring $\mathcal{R}$ has two units: "0" for addition and "1" for multiplication. The division rings commonly used are $\mathbb{R}$, $\mathbb{C}$ (commutative) and $\mathbb{H}$ (noncommutative). Commutative division rings are called fields. For commutative fields the inequality $A \circ B \neq B \circ A$ in the table turns into equality. All other properties remain unchanged. Quaternion division ring sometimes is referred to as non-commutative field, though this is not overwhelmingly accepted. In GA, therefore, $\mathbb{R}, \mathbb{C}$ and $\mathbb{H}$, collectively are called fields, or just $\mathbb{K}$ field. In the ideal theory the description of non-commutative GA rings produces a new GA structure of multivectors (MVs) named the spinor.

Ring algebra. If $A \in \mathcal{R}$ and requirement $AA^{-1} = A^{-1}A = 1$ is released, then the set of properties (axioms) listed in the Table I describes a noncommutative ring $\mathcal{R}$. The property of distributivity in this case splits into the left distributivity, $A \circ (B + C) = (A \circ B) + (A \circ C)$, and the right distributivity, $(B + C) \circ A = (B \circ A) + (C \circ A)$. The simplest example of noncommutative ring is a set of $n \times n$ real matrices. The algebra satisfies the same set of axioms as a ring. Formally, the algebra is a vector space $\mathcal{V}$ over a field $\mathbb{K}$ with a multiplication which turns it into a ring $\mathcal{R}$. The algebra also may have an additional definition, in particular, describing how to multiply an algebra element by a scalar (or external ring of scalars). An external ring of scalars in many cases, for example, in Clifford algebra, also may be internal, i.e., the scalars may belong to elements of algebra. A *division ring* or division algebra is a ring in which every nonzero element has a multiplicative inverse, but multiplication is not necessarily commutative. A commutative division algebra is also called a field.

The *left (right) ideal* $\mathcal{I}$ is a subset of the ring $\mathcal{R}$ which has an additional property: if $A \in \mathcal{I}$ is an ideal element and $B \in \mathcal{R}$ is any ring element then the noncommutative product $B \circ A \in \mathcal{I}$, (respectively $A \circ B \in \mathcal{I}$), i.e. belongs to left (right) ideal too. For example, the set of even integers is an ideal in the ring of all integers $\mathbb{Z}$, since any integer multiplied by even number is even. Also a matrix with one nonzero column (and other elements filled with zeroes) is left ideal element, since multiplication of it by arbitrary matrix from the left again gives a matrix with single column.

Name	Addition	Multiplication
Commutativity	$A + B = B + A$	$A \circ B \neq B \circ A$
Associativity	$(A + B) + C = A + (B + C)$	$(A \circ B) \circ C = A \circ (B \circ C)$
Distributivity	$A \circ (B + C) = (A \circ B) + (A \circ C)$	
Identity	$A + 0 = A = 0 + A$	$A \circ 1 = A = 1 \circ A$
Inverse	$A + (-A) = 0 = (-A) + A$	$AA^{-1} = 1 = A^{-1}A$

TABLE I: Properties of noncommutative division ring $\mathcal{R} \ni A, B, C$

II. ORDERING OF BASIS ELEMENTS IN GA

Tables III-XV lists the idempotent factors in the order of growing numbers in a blade index, exactly as letters are listed in the alphabet. Sometimes different ordering of basis elements is needed. In GA four main orderings of basis elements are defined: lexicographic (Lex), inverse lexicographic (InvLex), reverse lexicographic (RevLex), inverse reverse lexicographic (InvRevLex). These orderings may be preceded by two possible corrections "Deg" (higher degree blades first) or "InvDeg" (higher degree blades last), so all in all we have $4 \times 3 = 12$ possible orderings of GA basis elements. We will denote the additional orderings as Deg[InvLex], Deg[RevLex], Deg[InvRevLex], InvDeg[Lex], InvDeg[InvLex], InvDeg[RevLex] and InvDeg[InvRevLex]. To distinguish the named orderings an algebra of dimension $n \geq 4$ is required. Given two sets of indices (treated as components of abstract vector in the index space) $\alpha = \{\alpha_1, \alpha_2 \dots\}$ and $\beta = \{\beta_1, \beta_2 \dots\}$ with their lengths (which encode the grade of the element) denoted as $|\alpha|$ and $|\beta|$ the mentioned orderings are defined in the following way:

- Lex** ordering, $\alpha >_{\text{Lex}} \beta$, if in the vector difference $\alpha - \beta \in \mathbb{Z}^n$ the *left most* nonzero entry is *positive* (for index set with less elements zeroes are inserted in the place of absent index, see subsection II A below). An example (*Ex.*) of Lex ordering is
Ex. $\{\mathbf{e}_{1234}, \mathbf{e}_{123}, \mathbf{e}_{124}, \mathbf{e}_{12}, \mathbf{e}_{134}, \mathbf{e}_{13}, \mathbf{e}_{14}, \mathbf{e}_1, \mathbf{e}_{234}, \mathbf{e}_{23}, \mathbf{e}_{24}, \mathbf{e}_2, \mathbf{e}_{34}, \mathbf{e}_3, \mathbf{e}_4, 1\}$.
- InvLex** ordering, $\alpha >_{\text{InvLex}} \beta$, if in $\alpha - \beta \in \mathbb{Z}^n$ the *right most* nonzero entry is *positive*.
Ex. $\{\mathbf{e}_{1234}, \mathbf{e}_{234}, \mathbf{e}_{134}, \mathbf{e}_{34}, \mathbf{e}_{124}, \mathbf{e}_{24}, \mathbf{e}_{14}, \mathbf{e}_4, \mathbf{e}_{123}, \mathbf{e}_{23}, \mathbf{e}_{13}, \mathbf{e}_3, \mathbf{e}_{12}, \mathbf{e}_2, \mathbf{e}_1, 1\}$.
- RevLex** ordering, $\alpha >_{\text{RevLex}} \beta$, if in $\alpha - \beta \in \mathbb{Z}^n$ the *right most* nonzero entry is *negative*.
Ex. $\{1, \mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_{12}, \mathbf{e}_3, \mathbf{e}_{13}, \mathbf{e}_{23}, \mathbf{e}_{123}, \mathbf{e}_4, \mathbf{e}_{14}, \mathbf{e}_{24}, \mathbf{e}_{124}, \mathbf{e}_{34}, \mathbf{e}_{134}, \mathbf{e}_{234}, \mathbf{e}_{1234}\}$.
- InvRevLex** ordering, $\alpha >_{\text{InvRevLex}} \beta$, if in $\alpha - \beta \in \mathbb{Z}^n$ the *left most* nonzero entry is *negative*.
Ex. $\{1, \mathbf{e}_4, \mathbf{e}_3, \mathbf{e}_{34}, \mathbf{e}_2, \mathbf{e}_{24}, \mathbf{e}_{23}, \mathbf{e}_{234}, \mathbf{e}_1, \mathbf{e}_{14}, \mathbf{e}_{13}, \mathbf{e}_{134}, \mathbf{e}_{12}, \mathbf{e}_{124}, \mathbf{e}_{123}, \mathbf{e}_{1234}\}$.
- Deg[.]**
Lex ordering, $\alpha >_{\text{Deg[Lex]}} \beta$, if $|\alpha| > |\beta|$ and Lex when $|\alpha| = |\beta|$.
Ex. $\{\mathbf{e}_{1234}, \mathbf{e}_{123}, \mathbf{e}_{124}, \mathbf{e}_{134}, \mathbf{e}_{234}, \mathbf{e}_{12}, \mathbf{e}_{13}, \mathbf{e}_{14}, \mathbf{e}_{23}, \mathbf{e}_{24}, \mathbf{e}_{34}, \mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3, \mathbf{e}_4, 1\}$.
- InvLex** ordering, $\alpha >_{\text{Deg[InvLex]}} \beta$, if $|\alpha| > |\beta|$ and InvLex when $|\alpha| = |\beta|$.
Ex. $\{\mathbf{e}_{1234}, \mathbf{e}_{234}, \mathbf{e}_{134}, \mathbf{e}_{124}, \mathbf{e}_{123}, \mathbf{e}_{34}, \mathbf{e}_{24}, \mathbf{e}_{14}, \mathbf{e}_{23}, \mathbf{e}_{13}, \mathbf{e}_{12}, \mathbf{e}_4, \mathbf{e}_3, \mathbf{e}_2, \mathbf{e}_1, 1\}$.
- RevLex** ordering, $\alpha >_{\text{Deg[RevLex]}} \beta$, if $|\alpha| > |\beta|$ and RevLex when $|\alpha| = |\beta|$.
Ex. $\{\mathbf{e}_{1234}, \mathbf{e}_{123}, \mathbf{e}_{124}, \mathbf{e}_{134}, \mathbf{e}_{234}, \mathbf{e}_{12}, \mathbf{e}_{13}, \mathbf{e}_{23}, \mathbf{e}_{14}, \mathbf{e}_{24}, \mathbf{e}_{34}, \mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3, \mathbf{e}_4, 1\}$.
- InvRevLex** ordering, $\alpha >_{\text{Deg[InvRevLex]}} \beta$, if $|\alpha| > |\beta|$ and InvRevLex when $|\alpha| = |\beta|$.
Ex. $\{\mathbf{e}_{1234}, \mathbf{e}_{234}, \mathbf{e}_{134}, \mathbf{e}_{124}, \mathbf{e}_{123}, \mathbf{e}_{34}, \mathbf{e}_{24}, \mathbf{e}_{23}, \mathbf{e}_{14}, \mathbf{e}_{13}, \mathbf{e}_{12}, \mathbf{e}_4, \mathbf{e}_3, \mathbf{e}_2, \mathbf{e}_1, 1\}$.
- InvDeg[.]**
Lex ordering, $\alpha >_{\text{InvDeg[Lex]}} \beta$, if $|\alpha| < |\beta|$ and Lex when $|\alpha| = |\beta|$.
Ex. $\{1, \mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3, \mathbf{e}_4, \mathbf{e}_{12}, \mathbf{e}_{13}, \mathbf{e}_{14}, \mathbf{e}_{23}, \mathbf{e}_{24}, \mathbf{e}_{34}, \mathbf{e}_{123}, \mathbf{e}_{124}, \mathbf{e}_{134}, \mathbf{e}_{234}, \mathbf{e}_{1234}\}$
- InvLex** ordering, $\alpha >_{\text{InvDeg[InvLex]}} \beta$, if $|\alpha| < |\beta|$ and InvLex when $|\alpha| = |\beta|$.
Ex. $\{1, \mathbf{e}_4, \mathbf{e}_3, \mathbf{e}_2, \mathbf{e}_1, \mathbf{e}_{34}, \mathbf{e}_{24}, \mathbf{e}_{14}, \mathbf{e}_{23}, \mathbf{e}_{13}, \mathbf{e}_{12}, \mathbf{e}_{234}, \mathbf{e}_{134}, \mathbf{e}_{124}, \mathbf{e}_{123}, \mathbf{e}_{1234}\}$
- RevLex** ordering, $\alpha >_{\text{InvDeg[RevLex]}} \beta$, if $|\alpha| < |\beta|$ and RevLex when $|\alpha| = |\beta|$.
Ex. $\{1, \mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3, \mathbf{e}_4, \mathbf{e}_{12}, \mathbf{e}_{13}, \mathbf{e}_{23}, \mathbf{e}_{14}, \mathbf{e}_{24}, \mathbf{e}_{34}, \mathbf{e}_{123}, \mathbf{e}_{124}, \mathbf{e}_{134}, \mathbf{e}_{234}, \mathbf{e}_{1234}\}$
- InvRevLex** ordering, $\alpha >_{\text{InvDeg[InvRevLex]}} \beta$, if $|\alpha| < |\beta|$ and InvRevLex when $|\alpha| = |\beta|$.
Ex. $\{1, \mathbf{e}_4, \mathbf{e}_3, \mathbf{e}_2, \mathbf{e}_1, \mathbf{e}_{34}, \mathbf{e}_{24}, \mathbf{e}_{23}, \mathbf{e}_{14}, \mathbf{e}_{13}, \mathbf{e}_{12}, \mathbf{e}_{234}, \mathbf{e}_{134}, \mathbf{e}_{124}, \mathbf{e}_{123}, \mathbf{e}_{1234}\}$

Orderings RevLex, InvRevLex and the default ordering InvDeg[Lex] are not admissible as Gröbner basis monomial orderings. GA basis element ordering plays an important role in automatic computation of spinors. In particular if we want to construct a spinor, represented by a first matrix column we have to sort ideal basis elements in a specific order (see next section).

A. Example of blades $\mathbf{e}_{12}$ and $\mathbf{e}_{134}$ ordering

Given two basis elements $\mathbf{e}_{12}$ and $\mathbf{e}_{134}$ let's determine their position in all orderings. After insertion of zeroes we have $\{1, 2, 0, 0\} - \{1, 0, 3, 4\} = \{0, 2, -3, -4\}$. The first nonzero element in the *left most* position is 2. It is positive, therefore in Lex $\mathbf{e}_{12} >_{\text{Lex}} \mathbf{e}_{134}$. Similarly, for the same reason in InvLex order $\mathbf{e}_{12} <_{\text{InvLex}} \mathbf{e}_{134}$, because the *right most* element -4 is negative. For the same reason in RevLex $\mathbf{e}_{12} >_{\text{RevLex}} \mathbf{e}_{134}$. Finally, for InvRevLex the left most nonzero element is 2, therefore $\mathbf{e}_{12} <_{\text{InvRevLex}} \mathbf{e}_{134}$. It can be checked that the examples above incorporate both elements in the correct relative positions. Of course, in Deg[Lex], Deg[InvLex], Deg[RevLex] and Deg[InvRevLex] one has $\mathbf{e}_{12} <_{\text{Deg[]}} \mathbf{e}_{134}$ and in InvDeg[Lex], InvDeg[InvLex], InvDeg[RevLex] and InvDeg[InvRevLex] one has $\mathbf{e}_{12} >_{\text{InvDeg[]}} \mathbf{e}_{134}$.

III. REAL GA SPINORS AS MINIMAL LEFT IDEALS

The procedure to compute algebraic spinors and their matrix representations (reps) are described in detail in papers^{2,3}, that are adapted to *Mathematica* and further improved by adding automatic ordering of ideal basis elements⁴, spinors and their norms.

An algebraic spinor computation procedure is initiated by choosing a primitive idempotent. After a primitive idempotent of respective algebra is chosen one consecutively calculates the left ideals, two-sided ideals, left ideal basis (generators), respective matrix reps and finally the algebraic spinors. The obtained data are summarized in the Tables III-IX for real idempotents/spinors and Tables X-XV for complex idempotents/spinors. The items in the tables are enumerated from 1 to 7 and carry the following information:

1. defines a list of *mutually annihilating idempotents* $P_i \in Cl_{p,q}$ with factors sorted in $\text{InvDeg}(\text{Lex})$ order. The idempotents that differ by signs are singled out by plus/minus signs. For example, $P_1 = \frac{1}{2}(1 + \mathbf{e}_1) \equiv (+)$ and $P_2 = \frac{1}{2}(1 - \mathbf{e}_1) \equiv (-)$, or $P_1 = \frac{1}{4}(1 + \mathbf{e}_1)(1 + \mathbf{e}_{23}) \equiv (+, +)$ and $P_2 = \frac{1}{4}(1 + \mathbf{e}_1)(1 - \mathbf{e}_{23}) \equiv (+, -)$.
2. gives a *list of elements of two-sided ideal* $K_1 = \{K_1(1)\}$, or $K_1 = \{K_1(2)\}$, or $K_1 = \{K_1(4)\}$ generated by $K_1 = P_1\{Cl_{p,q}\}P_1$. The number of elements, denoted as (1),(2) or (4), in the list of two-sided ideal determines a type of matrix rep: $1 \leftrightarrow \mathbb{R}$, $2 \leftrightarrow \mathbb{C}$, and $4 \leftrightarrow \mathbb{H}$, i.e. if K_1 contains single element (elements that differ by sign are considered same element) which squares to +1 then the matrix rep entries are real numbers (matrix representation is real). When there are two different elements (one squares to +1, other to -1 and both commute), then the matrix representation entries are complex numbers (matrix representation is complex). Lastly, if there are four different non-zero elements then the matrix rep entries are quaternions and respective four elements $K(4)$ have to be replaced by quaternion units 1, q_1, q_2 , and q_3 by matching their properties.
3. provides *spinor basis* $S_1 = \{S_1(1), S_1(2), \dots\}P_1$ for the left ideal spinors $\mathcal{I}(P_1)$. The individual elements $S_1(i)$ are called the *projections* of the ideal $\mathcal{I}(P_1)$, or just *generators*. In quantum mechanics they represent *spinor basis* of a respective geometric algebra. The basis is sorted in RevLex order that ensures that the matrix rep of spinor will contain only the leftmost nonzero matrix column. The length of the ideal basis determines the dimension of the matrix rep (for simple algebras) and dimension of the block for the block-diagonal matrix (for semisimple algebras). In the latter case the remaining half of the ideal basis can be obtained by grade inversion and should be sorted in InvLex order to ensure proper place of the column of matrix representation of second block.
4. gives the matrix rep of basis vectors $\mathbf{e}_k$ in the minimal left ideal basis S_1 . The matrix type and dimension is in agreement with the 8-periodicity table¹. The symbol E_{ij} here denotes matrix with a single entry equal to 1 at the intersection of the i -th row and j -th column. For example, $E_{11} - E_{22} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$, or $q(-E_{12} + E_{21}) = \begin{bmatrix} 0 & -q \\ q & 0 \end{bmatrix}$, where q is the quaternion $q = q_0 + aq_1 + bq_2 + cq_3$ and $q_1^2 = q_2^2 = q_3^2 = -1$. The first of matrices is diagonal what indicates that the quantization axis of the spinor is parallel to GA basis vector $\mathbf{e}_1$ (this property is determined by choice of an idempotent in item 1.).
5. lists general spinor Ψ expanded in GA minimal left ideal basis, where s_i are the scalar expansion coefficients. In tables the spinor expressions for semisimple algebras includes two idempotents that are related by grade inversion (denoted by $\widehat{P}_i$).
6. gives the matrix rep $\hat{\Psi}$ of the GA general spinor Ψ .
7. expresses the square of the spinor norm $\|\Psi\|^2$, which is a positive scalar. The spinor hermitian norm can be written universally (in algebra independent way), as $\|\Psi\|^2 = \langle \Psi^\dagger \Psi \rangle$, where $\Psi^\dagger = \sum_T c_T^* \mathbf{e}_T^{-1}$ is the spinor in the reciprocal basis, an asterisk $*$ denotes complex conjugation of scalar coefficient and T is a multi-index, i.e. the sum is over all ordered orthonormal basis. The universal hermitian norm formula can be rewritten by algebra dependent combination of involutions presented in LHS of this item.

IV. SPINOR DATA TABLES (REAL IDEMPOTENTS)

The trivial primitive idempotents, $\{P_1 = 1\}$, in algebras $Cl_{0,1}$ and $Cl_{0,2}$ which, respectively, represent complex $\mathbb{C}$ and quaternion $\mathbb{H}$ numbers, suggest that the spinors in these algebras are division rings. In the tables below, $q_1 \equiv \mathbf{i}$, $q_2 \equiv \mathbf{j}$ and $q_3 \equiv \mathbf{k}$, where squares of quaternion units are $\mathbf{i}^2 = \mathbf{j}^2 = \mathbf{k}^2 = -1$. The arc over MV, for example $\widehat{\mathbf{e}_3 P_1}$, denotes grade inversion. The tilde $\widetilde{\Psi}$ represents reversion. The combination of both is a Clifford conjugation $\widehat{\widetilde{\Psi}}$.

	Item number and name	$Cl_{1,0}[\mathbb{R}(1)]$	$Cl_{0,1}[\mathbb{C}(1)]$
1	Idempotent, P_i	$\{P_1 = \frac{1}{2}(1 + \mathbf{e}_1), P_2 = (-) = \widehat{P_1}\}$	$\{P_1 = 1\}$
2	Two-sided ideal, K_1	$\{P_1\}$	$\{1, \mathbf{e}_1\}$
3	Projectors, (generators), S_1	$\{P_1\} \cup \{\widehat{P_1}\}$	$\{P_1\}$
4	Vector matrix rep, $\hat{e}_j$	$\{E_{11} - E_{22}\}$	$\{iE_{11}\}$
5	General spinor, Ψ	$s_1 P_1 + s_2 \widehat{P_1}$	$s_1 + s_2 \mathbf{e}_1$
6	Spinor in matrix form, $\hat{\Psi}$	$s_1 E_{11} + s_2 E_{22}$	$(s_1 + s_2 i) E_{11}$
7	Norm squared, $\ \Psi\ ^2$	$\langle \Psi \Psi \rangle = \frac{1}{2}(s_1^2 + s_2^2)$	$\widehat{\Psi} \Psi = s_1^2 + s_2^2$

TABLE III: Real spinor tables for GAs with vector space dimension $n = 1$

	$Cl_{2,0}[\mathbb{R}(2)]$	$Cl_{1,1}[\mathbb{R}(2)]$	$Cl_{0,2}[\mathbb{H}]$
1	$\{P_1 = \frac{1}{2}(1 + \mathbf{e}_1), P_2 = (-)\}$	$\{P_1 = \frac{1}{2}(1 + \mathbf{e}_1), P_2 = (-)\}$	$\{P_1 = 1\}$
2	$\{P_1\}$	$\{P_1\}$	$\{1, \mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$
3	$\{P_1, \mathbf{e}_2 P_1\}$	$\{P_1, \mathbf{e}_2 P_1\}$	$\{P_1\}$
4	$\{E_{11} - E_{22}, E_{12} + E_{21}\}$	$\{E_{11} - E_{22}, -E_{12} + E_{21}\}$	$\{q_1 E_{11}, q_2 E_{11}\}$
5	$\Psi = (s_1 + s_2 \mathbf{e}_2) P_1$	$\Psi = (s_1 + s_2 \mathbf{e}_2) P_1$	$\Psi = s_1 + s_2 \mathbf{e}_1 + s_3 \mathbf{e}_2 + s_4 \mathbf{e}_{12}$
6	$s_1 E_{11} + s_2 E_{12}$	$s_1 E_{11} + s_2 E_{12}$	$(s_1 + s_2 q_1 + s_3 q_2 + s_4 q_3) E_{11}$
7	$\langle \widetilde{\Psi} \Psi \rangle = \frac{1}{2}(s_1^2 + s_2^2)$	$\langle \mathbf{e}_1 \widetilde{\Psi} \mathbf{e}_1 \Psi \rangle = \frac{1}{2}(s_1^2 + s_2^2)$	$\widehat{\Psi} \Psi = s_1^2 + s_2^2 + s_3^2 + s_4^2$

TABLE IV: Real spinor table for GAs with vector space dimension $n = 2$

	$Cl_{3,0}[\mathbb{C}(2)]$	$Cl_{2,1}[\mathbb{R}(2)]$
1	$\{P_1 = \frac{1}{2}(1 + \mathbf{e}_1), P_2 = (-)\}$	$\{P_1 = \frac{1}{4}(1 + \mathbf{e}_1)(1 + \mathbf{e}_{23}), P_2 = (+, -), P_3 = (-, +), P_4 = (-, -)\}$
2	$\{P_1, P_1 P_1, P_1 \mathbf{e}_{23} P_1\}$	$\{P_1, P_1 P_1\}$
3	$\{P_1, \mathbf{e}_2 P_1\}$	$\{P_1, \mathbf{e}_2 P_1\} \cup \{\widehat{P_1}, \widehat{\mathbf{e}_2 P_1}\}$
4	$\{E_{11} - E_{22}, E_{12} + E_{21}, i(-E_{12} + E_{21})\}$	$\{E_{11} - E_{22} - E_{33} + E_{44}, E_{12} + E_{21} - E_{34} - E_{43}, -E_{12} + E_{21} + E_{34} - E_{43}\}$
5	$\Psi = (s_1 + s_2 \mathbf{e}_{23} + s_3 \mathbf{e}_2 - s_4 \mathbf{e}_3) P_1$	$\Psi = (s_1 + s_2 \mathbf{e}_2) P_1 + (s_3 - s_4 \mathbf{e}_2) \widehat{P_1}$
6	$(s_1 + i s_2) E_{11} + (s_3 - i s_4) E_{21}$	$s_1 E_{11} + s_2 E_{12} + s_3 E_{33} + s_4 E_{34}$
7	$\langle \widetilde{\Psi} \Psi \rangle = \frac{1}{2}(s_1^2 + \dots + s_4^2)$	$-\langle \mathbf{e}_3 \widetilde{\Psi} \mathbf{e}_3 \Psi \rangle = \frac{1}{4}(s_1^2 + \dots + s_4^2)$

	$Cl_{1,2}[\mathbb{C}(2)]$	$Cl_{0,3}[\mathbb{H}(2)]$
1	$\{P_1 = \frac{1}{2}(1 + \mathbf{e}_1), P_2 = (-)\}$	$\{P_1 = \frac{1}{2}(1 + \mathbf{e}_{123}), P_2 = \frac{1}{2}(1 - \mathbf{e}_{123})\}$
2	$\{P_1, P_1 P_1, P_1 \mathbf{e}_{23} P_1\}$	$\{P_1, P_1 P_1, P_1 \mathbf{e}_1 P_1, P_1 \mathbf{e}_2 P_1, P_1 \mathbf{e}_3 P_1\}$
3	$\{P_1, \mathbf{e}_2 P_1\}$	$\{\mathbf{e}_3 P_1\} \cup \{\widehat{\mathbf{e}_3 P_1}\}$
4	$\{E_{11} - E_{22}, -E_{12} + E_{21}, -iE_{12} - iE_{21}\}$	$\{-q_1 E_{11} + q_1 E_{22}, -q_2 E_{11} + q_2 E_{22}, -q_3 E_{11} + q_3 E_{22}\}$
5	$\Psi = (s_1 + s_2 \mathbf{e}_{23} + s_3 \mathbf{e}_2 + s_4 \mathbf{e}_3) P_1$	$\Psi = (-s_1 - s_2 \mathbf{e}_1 + s_3 \mathbf{e}_2 + s_4 \mathbf{e}_3) P_1 + (s_5 - s_6 \mathbf{e}_1 + s_7 \mathbf{e}_2 - s_8 \mathbf{e}_3) \widehat{P_1}$
6	$(s_1 + i s_2) E_{11} + (s_3 - i s_4) E_{21}$	$(-s_1 + s_2 q_1 - s_3 q_2 - s_4 q_3) E_{11} + (s_5 - s_6 q_1 + s_7 q_2 - s_8 q_3) E_{22}$
7	$\langle \mathbf{e}_1 \widetilde{\Psi} \mathbf{e}_1 \Psi \rangle = \frac{1}{2}(s_1^2 + \dots + s_4^2)$	$\langle \widetilde{\Psi} \Psi \rangle = \frac{1}{2}(s_1^2 + \dots + s_8^2)$

TABLE V: Real spinor table for GAs with vector space dimension $n = 3$

$Cl_{4,0}[\mathbb{H}(2)]$	$Cl_{3,1}[\mathbb{R}(4)]$
1 $\{P_1 = \frac{1}{2}(1 + \mathbf{e}_1), P_2 = \frac{1}{2}(1 - \mathbf{e}_1)\}$	$\{P_1 = \frac{1}{4}(1 + \mathbf{e}_1)(1 + \mathbf{e}_{24}), P_2 = (+, -), P_3 = (-, +), P_4 = (-, -)\}$
2 $\{P_1 1 P_1, P_1 \mathbf{e}_{23} P_1, P_1 \mathbf{e}_{24} P_1, P_1 \mathbf{e}_{34} P_1\}$	$\{P_1 1 P_1\}$
3 $\{P_1, \mathbf{e}_2 P_1\}$	$\{P_1, \mathbf{e}_2 P_1, \mathbf{e}_3 P_1, \mathbf{e}_{23} P_1\}$
4 $\{E_{11} - E_{22}, E_{12} + E_{21}, -q_1 E_{12} + q_1 E_{21},$ $-q_2 E_{12} + q_2 E_{21}\}$	$\{E_{11} - E_{22} - E_{33} + E_{44}, E_{12} + E_{21} + E_{34} + E_{43},$ $E_{13} - E_{24} + E_{31} - E_{42}, -E_{12} + E_{21} - E_{34} + E_{43}\}$
5 $\Psi = (s_1 + s_2 \mathbf{e}_{23} + s_3 \mathbf{e}_{24} + s_4 \mathbf{e}_{34} + s_5 \mathbf{e}_2 - s_6 \mathbf{e}_3$ $- s_7 \mathbf{e}_4 + s_8 \mathbf{e}_{234}) P_1$	$\Psi = (s_1 + s_2 \mathbf{e}_2 + s_3 \mathbf{e}_3 + s_4 \mathbf{e}_{23}) P_1$
6 $(s_1 + s_2 q_1 + s_3 q_2 - s_4 q_3) E_{11} + (s_5 - s_6 q_1 - s_7 q_2$ $- s_8 q_3) E_{21}$	$s_1 E_{11} + s_2 E_{21} + s_3 E_{31} + s_4 E_{41}$
7 $\langle \widetilde{\Psi} \Psi \rangle = \frac{1}{2}(s_1^2 + \dots + s_8^2)$	$-\langle \mathbf{e}_{123} \widetilde{\Psi} \mathbf{e}_{123} \Psi \rangle = \frac{1}{4}(s_1^2 + \dots + s_4^2)$
$Cl_{2,2}[\mathbb{R}(4)]$	$Cl_{1,3}[\mathbb{H}(2)]$
1 $\{P_1 = \frac{1}{4}(1 + \mathbf{e}_1)(1 + \mathbf{e}_{23}), P_2 = (+, -), P_3 = (-, +),$ $P_4 = (-, -)\}$	$\{P_1 = \frac{1}{2}(1 + \mathbf{e}_1), P_2 = \frac{1}{2}(1 - \mathbf{e}_1)\}$
2 $\{P_1 1 P_1\}$	$\{P_1 1 P_1, P_1 \mathbf{e}_{23} P_1, P_1 \mathbf{e}_{24} P_1, P_1 \mathbf{e}_{34} P_1\}$
3 $\{P_1, \mathbf{e}_2 P_1, \mathbf{e}_4 P_1, \mathbf{e}_{24} P_1\}$	$\{P_1, \mathbf{e}_2 P_1\}$
4 $\{E_{11} - E_{22} - E_{33} + E_{44}, E_{12} + E_{21} + E_{34} + E_{43},$ $-E_{12} + E_{21} - E_{34} + E_{43}, -E_{13} + E_{24} + E_{31} - E_{42}\}$	$\{E_{11} - E_{22}, -E_{12} + E_{21}, -q_1 E_{12} - q_1 E_{21}, -q_2 E_{12} - q_2 E_{21}\}$
5 $\Psi = (s_1 + s_2 \mathbf{e}_2 + s_3 \mathbf{e}_4 + s_4 \mathbf{e}_{24}) P_1$	$\Psi = (s_1 + s_2 \mathbf{e}_{23} + s_3 \mathbf{e}_{24} + s_4 \mathbf{e}_{34} + s_5 \mathbf{e}_2 + s_6 \mathbf{e}_3 + s_7 \mathbf{e}_4 + s_8 \mathbf{e}_{234}) P_1$
6 $s_1 E_{11} + s_2 E_{21} + s_3 E_{31} + s_4 E_{41}$	$(s_1 + s_2 q_1 + s_3 q_2 + s_4 q_3) E_{11} + (s_5 - s_6 q_1 - s_7 q_2 + s_8 q_3) E_{21}$
7 $-\langle \mathbf{e}_{34} \widetilde{\Psi} \mathbf{e}_{34} \Psi \rangle = \frac{1}{4}(s_1^2 + \dots + s_4^2)$	$\langle \mathbf{e}_1 \widetilde{\Psi} \mathbf{e}_1 \Psi \rangle = \frac{1}{2}(s_1^2 + \dots + s_8^2)$
$Cl_{0,4}[\mathbb{H}(2)]$	
1 $\{P_1 = \frac{1}{2}(1 + \mathbf{e}_{123}), P_2 = \frac{1}{2}(1 - \mathbf{e}_{123})\}$	
2 $\{P_1 1 P_1, P_1 \mathbf{e}_1 P_1, P_1 \mathbf{e}_2 P_1, P_1 \mathbf{e}_3 P_1\}$	
3 $\{\mathbf{e}_3 P_1, \mathbf{e}_{34} P_1\}$	
4 $\{-q_1 E_{11} + q_1 E_{22}, -q_2 E_{11} + q_2 E_{22}, -q_3 E_{11} + q_3 E_{22}, E_{12} - E_{21}\}$	
5 $\Psi = (-s_1 - s_2 \mathbf{e}_1 + s_3 \mathbf{e}_2 + s_4 \mathbf{e}_3 - s_5 \mathbf{e}_4 + s_6 \mathbf{e}_{1,4} - s_7 \mathbf{e}_{24} + s_8 \mathbf{e}_{34}) P_1$	
6 $(-s_1 + s_2 q_1 - s_3 q_2 - s_4 q_3) E_{11} + (s_5 - s_6 q_1 + s_7 q_2 - s_8 q_3) E_{21}$	
7 $\langle \widetilde{\Psi} \Psi \rangle = \frac{1}{2}(s_1^2 + \dots + s_8^2)$	

TABLE VI: Real spinor tables for GAs with vector space dimension $n = 4$

$Cl_{5,0}[\mathbb{H}(2)]$	$Cl_{4,1}[\mathbb{C}(4)]$
1 $\{P_1 = \frac{1}{4}(1 + \mathbf{e}_1)(1 + \mathbf{e}_{2345}), P_2 = \frac{1}{4}(1 + \mathbf{e}_1)(1 - \mathbf{e}_{2345}),$ $P_3 = \widehat{P_1} = \frac{1}{4}(1 - \mathbf{e}_1)(1 + \mathbf{e}_{2345}), P_4 = \frac{1}{4}(1 - \mathbf{e}_1)(1 - \mathbf{e}_{2345})\}$	$\{P_1 = \frac{1}{4}(1 + \mathbf{e}_1)(1 + \mathbf{e}_{25}), P_2 = \frac{1}{4}(1 + \mathbf{e}_1)(1 - \mathbf{e}_{25}),$ $P_3 = \frac{1}{4}(1 - \mathbf{e}_1)(1 + \mathbf{e}_{25}), P_4 = \frac{1}{4}(1 - \mathbf{e}_1)(1 - \mathbf{e}_{25})\}$
2 $\{P_1 P_1, P_1 \mathbf{e}_{23} P_1, P_1 \mathbf{e}_{24} P_1, P_1 \mathbf{e}_{25} P_1\}$	$\{P_1 P_1, P_1 \mathbf{e}_{34} P_1\}$
3 $\{\mathbf{e}_{25} P_1, \mathbf{e}_5 P_1\} \cup \{\widehat{\mathbf{e}_{25} P_1}, \widehat{\mathbf{e}_5 P_1}\}$	$\{P_1, \mathbf{e}_2 P_1, \mathbf{e}_3 P_1, \mathbf{e}_{23} P_1\}$
4 $\{E_{11} - E_{22} - E_{33} + E_{44}, E_{12} + E_{21} - E_{34} - E_{43},$ $q_1(E_{12} - E_{21} - E_{34} + E_{43}), q_2(E_{12} - E_{21} - E_{34} + E_{43}),$ $q_3(-E_{12} + E_{21} + E_{34} - E_{43})\}$	$\{E_{11} - E_{22} - E_{33} + E_{44}, E_{12} + E_{21} + E_{34} + E_{43},$ $E_{13} - E_{24} + E_{31} - E_{42}, i(-E_{13} + E_{24} + E_{31} - E_{42}),$ $-E_{12} + E_{21} - E_{34} + E_{43}\}$
5 $\Psi = (-s_1 + s_2 \mathbf{e}_{23} - s_3 \mathbf{e}_{24} + s_4 \mathbf{e}_{25} + s_5 \mathbf{e}_2 - s_6 \mathbf{e}_3 + s_7 \mathbf{e}_4 + s_8 \mathbf{e}_5) P_1$ $+ (-s_9 + s_{10} \mathbf{e}_{23} - s_{11} \mathbf{e}_{24} + s_{12} \mathbf{e}_{25} - s_{13} \mathbf{e}_2$ $+ s_{14} \mathbf{e}_3 - s_{15} \mathbf{e}_4 - s_{16} \mathbf{e}_5) \widehat{P_1}$	$\Psi = (s_1 + s_2 \mathbf{e}_{34} + s_3 \mathbf{e}_2 + s_4 \mathbf{e}_{234}$ $+ s_5 \mathbf{e}_3 - s_6 \mathbf{e}_4 + s_7 \mathbf{e}_{23} - s_8 \mathbf{e}_{24}) P_1$
6 $(-s_1 - s_2 q_1 + s_3 q_2 + s_4 q_3) E_{11} + (s_5 + s_6 q_1 - s_7 q_2 + s_8 q_3) E_{21}$ $+ (-s_9 - s_{10} q_1 + s_{11} q_2 + s_{12} q_3) E_{33}$ $+ (s_{13} + s_{14} q_1 - s_{15} q_2 + s_{16} q_3) E_{43}$	$(s_1 + i s_2) E_{11} + (s_3 + i s_4) E_{21} + (s_5 - i s_6) E_{31}$ $+ (s_7 - i s_8) E_{41}$
7 $\langle \widetilde{\Psi} \Psi \rangle = \frac{1}{4}(s_1^2 + \dots + s_{16}^2)$	$-\langle \mathbf{e}_5 \widetilde{\Psi} \mathbf{e}_5 \Psi \rangle = \frac{1}{4}(s_1^2 + \dots + s_8^2)$
$Cl_{3,2}[\mathbb{R}(4)]$	
1 $\{P_1 = \frac{1}{8}(1 + \mathbf{e}_1)(1 + \mathbf{e}_{24})(1 + \mathbf{e}_{35}), P_2 = (+, +, -), P_3 = (+, -, +), P_4 = (+, -, -), P_5 = \widehat{P_1} = (-, +, +), P_6 = (-, +, -),$ $P_7 = (-, -, +), P_8 = (-, -, -)\}$	
2 $\{P_1 P_1\}$	
3 $\{P_1, \mathbf{e}_2 P_1, \mathbf{e}_3 P_1, \mathbf{e}_{23} P_1\} \cup \{\widehat{P_1}, \widehat{\mathbf{e}_2 P_1}, \widehat{\mathbf{e}_3 P_1}, \widehat{\mathbf{e}_{23} P_1}\}$	
4 $\{E_{11} - E_{22} - E_{33} + E_{44} - E_{55} + E_{66} + E_{77} - E_{88}, E_{12} + E_{21} + E_{34} + E_{43} - E_{56} - E_{65} - E_{78} - E_{87},$ $E_{13} - E_{24} + E_{31} - E_{42} - E_{57} + E_{68} - E_{75} + E_{86}, -E_{12} + E_{21} - E_{34} + E_{43} + E_{56} - E_{65} + E_{78} - E_{87},$ $-E_{13} + E_{24} + E_{31} - E_{42} + E_{57} - E_{68} - E_{75} + E_{86}\}$	
5 $\Psi = (s_1 + s_2 \mathbf{e}_2 + s_3 \mathbf{e}_3 + s_4 \mathbf{e}_{23}) P_1 + (s_5 - s_6 \mathbf{e}_2 - s_7 \mathbf{e}_3 + s_8 \mathbf{e}_{23}) \widehat{P_1}$	
6 $s_1 E_{11} + s_2 E_{21} + s_3 E_{31} + s_4 E_{41} + s_5 E_{55} + s_6 E_{65} + s_7 E_{75} + s_8 E_{85}$	
7 $-\langle \mathbf{e}_{45} \widetilde{\Psi} \mathbf{e}_{45} \Psi \rangle = \frac{1}{8}(s_1^2 + \dots + s_8^2)$	
$Cl_{2,3}[\mathbb{C}(4)]$	
1 $\{P_1 = \frac{1}{4}(1 + \mathbf{e}_1)(1 + \mathbf{e}_{23}), P_2 = (+, -), P_3 = (-, +), P_4 = (-, -)\}$	
2 $\{P_1 P_1, P_1 \mathbf{e}_{45} P_1\}$	
3 $\{P_1, \mathbf{e}_2 P_1, \mathbf{e}_4 P_1, \mathbf{e}_{24} P_1\}$	
4 $\{E_{11} - E_{22} - E_{33} + E_{44}, E_{12} + E_{21} + E_{34} + E_{43}, -E_{12} + E_{21} - E_{34} + E_{43}, -E_{13} + E_{24} + E_{31} - E_{42},$ $i(-E_{13} + E_{24} - E_{31} + E_{42})\}$	
5 $\Psi = (s_1 + s_2 \mathbf{e}_{45} + s_3 \mathbf{e}_2 + s_4 \mathbf{e}_{245} + s_5 \mathbf{e}_4 + s_6 \mathbf{e}_5 + s_7 \mathbf{e}_{24} + s_8 \mathbf{e}_{25}) P_1$	
6 $(s_1 + i s_2) E_{11} + (s_3 + i s_4) E_{21} + (s_5 - i s_6) E_{31} + (s_7 - i s_8) E_{41}$	
7 $-\langle \mathbf{e}_{12} \widetilde{\Psi} \mathbf{e}_{12} \Psi \rangle = \frac{1}{8}(s_1^2 + \dots + s_8^2)$	
$Cl_{1,4}[\mathbb{H}(2)]$	
1 $\{P_1 = \frac{1}{4}(1 + \mathbf{e}_1)(1 + \mathbf{e}_{2345}), P_2 = (+, -), P_3 = \widehat{P_1} = (-, +), P_4 = (-, -)\}$	
2 $\{P_1 P_1, P_1 \mathbf{e}_{23} P_1, P_1 \mathbf{e}_{24} P_1, P_1 \mathbf{e}_{25} P_1\}$	
3 $\{\mathbf{e}_{25} P_1, \mathbf{e}_5 P_1\} \cup \{\widehat{\mathbf{e}_{25} P_1}, \widehat{\mathbf{e}_5 P_1}\}$	
4 $\{E_{11} - E_{22} - E_{33} + E_{44}, E_{12} - E_{21} - E_{34} + E_{43}, q_1(-E_{12} - E_{21} + E_{34} + E_{43}), q_2(-E_{12} - E_{21} + E_{34} + E_{43}),$ $q_3(-E_{12} - E_{21} + E_{34} + E_{43})\}$	
5 $\Psi = (-s_1 - s_2 \mathbf{e}_{23} + s_3 \mathbf{e}_{24} + s_4 \mathbf{e}_{25} - s_5 \mathbf{e}_2 + s_6 \mathbf{e}_3 - s_7 \mathbf{e}_4 + s_8 \mathbf{e}_5) P_1$ $+ (-s_9 - s_{10} \mathbf{e}_{23} + s_{11} \mathbf{e}_{24} + s_{12} \mathbf{e}_{25} + s_{13} \mathbf{e}_2 - s_{14} \mathbf{e}_3 + s_{15} \mathbf{e}_4 - s_{16} \mathbf{e}_5) \widehat{P_1}$	
6 $(-s_1 + s_2 q_1 - s_3 q_2 - s_4 q_3) E_{11} + (s_5 - s_6 q_1 + s_7 q_2 - s_8 q_3) E_{21} + (-s_9 + s_{10} q_1 - s_{11} q_2 - s_{12} q_3) E_{33}$ $+ (s_{13} - s_{14} q_1 + s_{15} q_2 - s_{16} q_3) E_{43}$	
7 $\langle \mathbf{e}_1 \widetilde{\Psi} \mathbf{e}_1 \Psi \rangle = \frac{1}{8}(s_1^2 + \dots + s_{16}^2)$	
$Cl_{0,5}[\mathbb{C}(4)]$	
1 $\{P_1 = \frac{1}{4}(1 + \mathbf{e}_{123})(1 + \mathbf{e}_{145}), P_2 = (+, -), P_3 = (-, +), P_4 = (-, -)\}$	
2 $\{P_1 P_1, P_1 \mathbf{e}_1 P_1\}$	
3 $\{P_1, \mathbf{e}_2 P_1, \mathbf{e}_4 P_1, \mathbf{e}_{24} P_1\}$	
4 $\{i(E_{11} - E_{22} - E_{33} + E_{44}), -E_{12} + E_{21} - E_{34} + E_{43}, i(E_{12} + E_{21} + E_{34} + E_{43}), -E_{13} + E_{24} + E_{31} - E_{42},$ $i(E_{13} - E_{24} + E_{31} - E_{42})\}$	
5 $\Psi = (s_1 + s_2 \mathbf{e}_1 + s_3 \mathbf{e}_2 - s_4 \mathbf{e}_3 + s_5 \mathbf{e}_4 - s_6 \mathbf{e}_5 + s_7 \mathbf{e}_{24} + s_8 \mathbf{e}_{25}) P_1$	
6 $(s_1 + i s_2) E_{11} + (s_3 - i s_4) E_{21} + (s_5 - i s_6) E_{31} + (s_7 + i s_8) E_{41}$	
7 $\langle \widetilde{\Psi} \Psi \rangle = \frac{1}{4}(s_1^2 + \dots + s_8^2)$	

TABLE VII: Real spinor tables for GAs with vector space dimension $n = 5$

$Cl_{6,0}[\mathbb{H}(4)]$	
1	$\{P_1 = \frac{1}{4}(1 + \mathbf{e}_1)(1 + \mathbf{e}_{2345}), P_2 = (+, -), P_3 = \widetilde{P_1} = (-, +), P_4 = (-, -)\}$
2	$\{P_1 P_1, P_1 \mathbf{e}_{23} P_1, P_1 \mathbf{e}_{24} P_1, P_1 \mathbf{e}_{25} P_1\}$
3	$\{\mathbf{e}_{25} P_1, \mathbf{e}_5 P_1, \mathbf{e}_{256} P_1, \mathbf{e}_{56} P_1\}$
4	$\{E_{11} - E_{22} - E_{33} + E_{44}, E_{12} + E_{21} + E_{34} + E_{43}, q_1(E_{12} - E_{21} + E_{34} - E_{43}), q_2(E_{12} - E_{21} + E_{34} - E_{43}),$ $q_3(-E_{12} + E_{21} - E_{34} + E_{43}), E_{13} - E_{24} + E_{31} - E_{42}\}$
5	$\Psi = (-s_1 + s_2 \mathbf{e}_{23} - s_3 \mathbf{e}_{24} + s_4 \mathbf{e}_{25} + s_5 \mathbf{e}_2 - s_6 \mathbf{e}_3 + s_7 \mathbf{e}_4 + s_8 \mathbf{e}_5$ $- s_9 \mathbf{e}_6 + s_{10} \mathbf{e}_{236} - s_{11} \mathbf{e}_{246} + s_{12} \mathbf{e}_{256} + s_{13} \mathbf{e}_{26} - s_{14} \mathbf{e}_{36} + s_{15} \mathbf{e}_{46} + s_{16} \mathbf{e}_{56}) P_1$
6	$(-s_1 - s_2 q_1 + s_3 q_2 + s_4 q_3) E_{11} + (s_5 + s_6 q_1 - s_7 q_2 + s_8 q_3) E_{21} + (-s_9 - s_{10} q_1 + s_{11} q_2 + s_{12} q_3) E_{31}$ $+ (s_{13} + s_{14} q_1 - s_{15} q_2 + s_{16} q_3) E_{41}$
7	$\langle \widetilde{\Psi} \Psi \rangle = \frac{1}{4}(s_1^2 + \dots + s_{16}^2)$
$Cl_{5,1}[\mathbb{H}(4)]$	
1	$\{P_1 = \frac{1}{4}(1 + \mathbf{e}_1)(1 + \mathbf{e}_{26}), P_2 = (+, -), P_3 = (-, +), P_4 = (-, -)\}$
2	$\{P_1 P_1, P_1 \mathbf{e}_{34} P_1, P_1 \mathbf{e}_{35} P_1, P_1 \mathbf{e}_{45} P_1\}$
3	$\{P_1, \mathbf{e}_2 P_1, \mathbf{e}_3 P_1, \mathbf{e}_{23} P_1\}$
4	$\{E_{11} - E_{22} - E_{33} + E_{44}, E_{12} + E_{21} + E_{34} + E_{43}, E_{13} - E_{24} + E_{31} - E_{42}, q_1(-E_{13} + E_{24} + E_{31} - E_{42}),$ $q_2(-E_{13} + E_{24} + E_{31} - E_{42}), -E_{12} + E_{21} - E_{34} + E_{43}\}$
5	$\Psi = (s_1 + s_2 \mathbf{e}_{34} + s_3 \mathbf{e}_{35} + s_4 \mathbf{e}_{45} + s_5 \mathbf{e}_2 + s_6 \mathbf{e}_{234} + s_7 \mathbf{e}_{235} + s_8 \mathbf{e}_{245}$ $+ s_9 \mathbf{e}_3 - s_{10} \mathbf{e}_4 - s_{11} \mathbf{e}_5 + s_{12} \mathbf{e}_{345} + s_{13} \mathbf{e}_{23} - s_{14} \mathbf{e}_{24} - s_{15} \mathbf{e}_{25} + s_{16} \mathbf{e}_{2345}) P_1$
6	$(s_1 + s_2 q_1 + s_3 q_2 - s_4 q_3) E_{11} + (s_5 + s_6 q_1 + s_7 q_2 - s_8 q_3) E_{21} + (s_9 - s_{10} q_1 - s_{11} q_2 - s_{12} q_3) E_{31}$ $+ (s_{13} - s_{14} q_1 - s_{15} q_2 - s_{16} q_3) E_{41}$
7	$-\langle \widetilde{\mathbf{e}_6} \Psi \mathbf{e}_6 \Psi \rangle = \frac{1}{4}(s_1^2 + \dots + s_{16}^2)$
$Cl_{4,2}[\mathbb{R}(8)]$	
1	$\{P_1 = \frac{1}{8}(1 + \mathbf{e}_1)(1 + \mathbf{e}_{25})(1 + \mathbf{e}_{36}), P_2 = (+, +, -), P_3 = (+, -, +), P_4 = (+, -, -), P_5 = (-, +, +), P_6 = (-, +, -),$ $P_7 = (-, -, +), P_8 = (-, -, -)\}$
2	$\{P_1 P_1\}$
3	$\{P_1, \mathbf{e}_2 P_1, \mathbf{e}_3 P_1, \mathbf{e}_{23} P_1, \mathbf{e}_4 P_1, \mathbf{e}_{24} P_1, \mathbf{e}_{34} P_1, \mathbf{e}_{234} P_1\}$
4	$\{E_{11} - E_{22} - E_{33} + E_{44} - E_{55} + E_{66} + E_{77} - E_{88}, E_{12} + E_{21} + E_{34} + E_{43} + E_{56} + E_{65} + E_{78} + E_{87},$ $E_{13} - E_{24} + E_{31} - E_{42} + E_{57} - E_{68} + E_{75} - E_{86}, E_{15} - E_{26} - E_{37} + E_{48} + E_{51} - E_{62} - E_{73} + E_{84},$ $-E_{12} + E_{21} - E_{34} + E_{43} - E_{56} + E_{65} - E_{78} + E_{87}, -E_{13} + E_{24} + E_{31} - E_{42} - E_{57} + E_{68} + E_{75} - E_{86}\}$
5	$\Psi = (s_1 + s_2 \mathbf{e}_2 + s_3 \mathbf{e}_3 + s_4 \mathbf{e}_{23} + s_5 \mathbf{e}_4 + s_6 \mathbf{e}_{24} + s_7 \mathbf{e}_{34} + s_8 \mathbf{e}_{234}) P_1$
6	$s_1 E_{11} + s_2 E_{21} + s_3 E_{31} + s_4 E_{41} + s_5 E_{51} + s_6 E_{61} + s_7 E_{71} + s_8 E_{81}$
7	$-\langle \widetilde{\mathbf{e}_{56}} \Psi \mathbf{e}_{56} \Psi \rangle = \frac{1}{8}(s_1^2 + \dots + s_8^2)$
$Cl_{3,3}[\mathbb{R}(8)]$	
1	$\{P_1 = \frac{1}{8}(1 + \mathbf{e}_1)(1 + \mathbf{e}_{24})(1 + \mathbf{e}_{35}), P_2 = (+, +, -), P_3 = (+, -, +), P_4 = (+, -, -), P_5 = (-, +, +), P_6 = (-, +, -),$ $P_7 = (-, -, +), P_8 = (-, -, -)\}$
2	$\{P_1 P_1\}$
3	$\{P_1, \mathbf{e}_2 P_1, \mathbf{e}_3 P_1, \mathbf{e}_{23} P_1, \mathbf{e}_6 P_1, \mathbf{e}_{26} P_1, \mathbf{e}_{36} P_1, \mathbf{e}_{236} P_1\}$
4	$\{E_{11} - E_{22} - E_{33} + E_{44} - E_{55} + E_{66} + E_{77} - E_{88}, E_{12} + E_{21} + E_{34} + E_{43} + E_{56} + E_{65} + E_{78} + E_{87},$ $E_{13} - E_{24} + E_{31} - E_{42} + E_{57} - E_{68} + E_{75} - E_{86}, -E_{12} + E_{21} - E_{34} + E_{43} - E_{56} + E_{65} - E_{78} + E_{87},$ $-E_{13} + E_{24} + E_{31} - E_{42} - E_{57} + E_{68} + E_{75} - E_{86}, -E_{15} + E_{26} + E_{37} - E_{48} + E_{51} - E_{62} - E_{73} + E_{84}\}$
5	$\Psi = (s_1 + s_2 \mathbf{e}_2 + s_3 \mathbf{e}_3 + s_4 \mathbf{e}_{23} + s_5 \mathbf{e}_6 + s_6 \mathbf{e}_{26} + s_7 \mathbf{e}_{36} + s_8 \mathbf{e}_{236}) P_1$
6	$s_1 E_{11} + s_2 E_{21} + s_3 E_{31} + s_4 E_{41} + s_5 E_{51} + s_6 E_{61} + s_7 E_{71} + s_8 E_{81}$
7	$-\langle \mathbf{e}_{123} \widetilde{\Psi} \mathbf{e}_{123} \Psi \rangle = \frac{1}{8}(s_1^2 + \dots + s_8^2)$
$Cl_{2,4}[\mathbb{H}(4)]$	
1	$\{P_1 = \frac{1}{4}(1 + \mathbf{e}_1)(1 + \mathbf{e}_{23}), P_2 = (+, -), P_3 = (-, +), P_4 = (-, -)\}$
2	$\{P_1 P_1, P_1 \mathbf{e}_{45} P_1, P_1 \mathbf{e}_{46} P_1, P_1 \mathbf{e}_{56} P_1\}$
3	$\{P_1, \mathbf{e}_2 P_1, \mathbf{e}_4 P_1, \mathbf{e}_{24} P_1\}$
4	$\{E_{11} - E_{22} - E_{33} + E_{44}, E_{12} + E_{21} + E_{34} + E_{43}, -E_{12} + E_{21} - E_{34} + E_{43}, -E_{13} + E_{24} + E_{31} - E_{42},$ $q_1(-E_{13} + E_{24} - E_{31} + E_{42}), q_2(-E_{13} + E_{24} - E_{31} + E_{42})\}$
5	$\Psi = (s_1 + s_2 \mathbf{e}_{45} + s_3 \mathbf{e}_{46} + s_4 \mathbf{e}_{56} + s_5 \mathbf{e}_2 + s_6 \mathbf{e}_{245} + s_7 \mathbf{e}_{246} + s_8 \mathbf{e}_{256} + s_9 \mathbf{e}_4 + s_{10} \mathbf{e}_5 + s_{11} \mathbf{e}_6 + s_{12} \mathbf{e}_{456} + s_{13} \mathbf{e}_{24} + s_{14} \mathbf{e}_{25}$ $+ s_{15} \mathbf{e}_{26} + s_{16} \mathbf{e}_{2456}) P_1$
6	$(s_1 + s_2 q_1 + s_3 q_2 + s_4 q_3) E_{11} + (s_5 + s_6 q_1 + s_7 q_2 + s_8 q_3) E_{21}$ $+ (s_9 - s_{10} q_1 - s_{11} q_2 + s_{12} q_3) E_{31} + (s_{13} - s_{14} q_1 - s_{15} q_2 + s_{16} q_3) E_{41}$
7	$-\langle \mathbf{e}_{12} \widetilde{\Psi} \mathbf{e}_{12} \Psi \rangle = \frac{1}{4}(s_1^2 + \dots + s_{16}^2)$

TABLE VIII: Real spinor tables for GAs $Cl_{6,0}$, $Cl_{5,1}$, $Cl_{4,2}$, $Cl_{3,3}$, and $Cl_{2,4}$; $n = 6$

$Cl_{1,5}[\mathbb{H}(4)]$	
1	$\{P_1 = \frac{1}{4}(1 + \mathbf{e}_1)(1 + \mathbf{e}_{2345}), P_2 = (+, -), P_3 = (-, +), P_4 = (-, -)\}$
2	$\{P_1 P_1, P_1 \mathbf{e}_{23} P_1, P_1 \mathbf{e}_{24} P_1, P_1 \mathbf{e}_{25} P_1\}$
3	$\{\mathbf{e}_{25} P_1, \mathbf{e}_5 P_1, \mathbf{e}_{256} P_1, \mathbf{e}_{56} P_1\}$
4	$\{E_{11} - E_{22} - E_{33} + E_{44}, E_{12} - E_{21} + E_{34} - E_{43}, q_1(-E_{12} - E_{21} - E_{34} - E_{43}), q_2(-E_{12} - E_{21} - E_{34} - E_{43}),$ $q_3(-E_{12} - E_{21} - E_{34} - E_{43}), -E_{13} + E_{24} + E_{31} - E_{42}\}$
5	$\Psi = (-s_1 - s_2 \mathbf{e}_{23} + s_3 \mathbf{e}_{24} + s_4 \mathbf{e}_{25} - s_5 \mathbf{e}_2 + s_6 \mathbf{e}_3 - s_7 \mathbf{e}_4 + s_8 \mathbf{e}_5 - s_9 \mathbf{e}_6 - s_{10} \mathbf{e}_{236} + s_{11} \mathbf{e}_{246} + s_{12} \mathbf{e}_{256} - s_{13} \mathbf{e}_{26} + s_{14} \mathbf{e}_{36}$ $- s_{15} \mathbf{e}_{46} + s_{16} \mathbf{e}_{56}) P_1$
6	$(-s_1 + s_2 q_1 - s_3 q_2 - s_4 q_3) E_{11} + (s_5 - s_6 q_1 + s_7 q_2 - s_8 q_3) E_{21}$ $+ (-s_9 + s_{10} q_1 - s_{11} q_2 - s_{12} q_3) E_{31} + (s_{13} - s_{14} q_1 + s_{15} q_2 - s_{16} q_3) E_{41}$
7	$\langle \mathbf{e}_1 \widetilde{\Psi} \mathbf{e}_1 \Psi \rangle = \frac{1}{4}(s_1^2 + \dots + s_{16}^2)$
$Cl_{0,6}[\mathbb{R}(8)]$	
1	$\{P_1 = \frac{1}{8}(1 + \mathbf{e}_{123})(1 + \mathbf{e}_{145})(1 + \mathbf{e}_{246}), P_2 = (+, +, -), P_3 = (+, -, +), P_4 = (+, -, -), P_5 = (-, +, +), P_6 = (-, +, -),$ $P_7 = (-, -, +), P_8 = (-, -, -)\}$
2	$\{P_1 P_1\}$
3	$\{P_1, \mathbf{e}_1 P_1, \mathbf{e}_2 P_1, \mathbf{e}_3 P_1, \mathbf{e}_4 P_1, \mathbf{e}_5 P_1, \mathbf{e}_6 P_1, \mathbf{e}_{16} P_1\}$
4	$\{-E_{12} + E_{21} + E_{34} - E_{43} + E_{56} - E_{65} - E_{78} + E_{87}, -E_{13} - E_{24} + E_{31} + E_{42} + E_{57} + E_{68} - E_{75} - E_{86},$ $-E_{14} + E_{23} - E_{32} + E_{41} + E_{58} - E_{67} + E_{76} - E_{85}, -E_{15} - E_{26} - E_{37} - E_{48} + E_{51} + E_{62} + E_{73} + E_{84},$ $-E_{16} + E_{25} - E_{38} + E_{47} - E_{52} + E_{61} - E_{74} + E_{83}, -E_{17} + E_{28} + E_{35} - E_{46} - E_{53} + E_{64} + E_{71} - E_{82}\}$
5	$\Psi = (s_1 + s_2 \mathbf{e}_1 + s_3 \mathbf{e}_2 + s_4 \mathbf{e}_3 + s_5 \mathbf{e}_4 + s_6 \mathbf{e}_5 + s_7 \mathbf{e}_6 + s_8 \mathbf{e}_{16}) P_1$
6	$s_1 E_{11} + s_2 E_{21} + s_3 E_{31} + s_4 E_{41} + s_5 E_{51} + s_6 E_{61} + s_7 E_{71} + s_8 E_{81}$
7	$\langle \widetilde{\Psi} \Psi \rangle = \frac{1}{8}(s_1^2 + \dots + s_8^2)$

TABLE IX: Real spinor tables for GAs $Cl_{1,5}$ and $Cl_{0,6}$; $n = 6$

A. Example of construction of the real spinor data table for $Cl_{2,2}$ algebra

1. *Primitive idempotents P .* In $Cl_{2,2}$ we need to choose a single pair of mutually commuting blades $(\mathbf{e}_{T_1}, \mathbf{e}_{T_2})$,

$$\{(\mathbf{e}_1, \mathbf{e}_{23}), (\mathbf{e}_1, \mathbf{e}_{24}), (\mathbf{e}_1, \mathbf{e}_{123}), (\mathbf{e}_1, \mathbf{e}_{124}), (\mathbf{e}_2, \mathbf{e}_{13}), (\mathbf{e}_2, \mathbf{e}_{14}), (\mathbf{e}_2, \mathbf{e}_{123}), (\mathbf{e}_2, \mathbf{e}_{124}), (\mathbf{e}_{13}, \mathbf{e}_{24}), (\mathbf{e}_{13}, \mathbf{e}_{123}),$$

$$(\mathbf{e}_{13}, \mathbf{e}_{1234}), (\mathbf{e}_{14}, \mathbf{e}_{23}), (\mathbf{e}_{14}, \mathbf{e}_{124}), (\mathbf{e}_{14}, \mathbf{e}_{1234}), (\mathbf{e}_{23}, \mathbf{e}_{123}), (\mathbf{e}_{23}, \mathbf{e}_{1234}), (\mathbf{e}_{24}, \mathbf{e}_{124}), (\mathbf{e}_{24}, \mathbf{e}_{1234})\}$$

ordered in $\text{InvDeg}[\text{Lex}]$ that all squares to $+1$ and can be used in starting the construction of a spinor (see Table VI). Each pair is mutually annihilating: $P_i P_j = P_j P_i = 0$ if $i \neq j$, and $i, j = 1, 2$. From all possible sets we always will use the first one in the $\text{InvDeg}[\text{Lex}]$ list, which means that we usually choose $\mathbf{e}_1$ direction as a quantization axis. Exceptions are anti-euclidean algebras $Cl_{0,q}$. In accord with Table VI we choose the pair, $(\mathbf{e}_1, \mathbf{e}_{23})$ and the primitive idempotent $P_1 \rightarrow P = \frac{1}{4}(1 + \mathbf{e}_1)(1 + \mathbf{e}_{23}) = \frac{1}{4}(1 + \mathbf{e}_1 + \mathbf{e}_{23} + \mathbf{e}_{123}) \equiv (+, +)$.

Once the idempotent P is chosen, multiply all basis elements of $Cl_{2,2}$ by P from right, $S = \{Cl_{2,2}\}P$. Elements that differ by sign will be treated as equivalent, therefore, only a single copy (positive) is used in the following. One gets that the left ideal S consists of four different members, $S = \{P, \mathbf{e}_2 P, \mathbf{e}_4 P, \mathbf{e}_{24} P\} = \{\frac{1}{4}(1 + \mathbf{e}_1 + \mathbf{e}_{23} + \mathbf{e}_{123}), \frac{1}{4}(\mathbf{e}_2 + \mathbf{e}_3 - \mathbf{e}_{12} - \mathbf{e}_{13}), \frac{1}{4}(\mathbf{e}_4 - \mathbf{e}_{14} + \mathbf{e}_{234} - \mathbf{e}_{1234}), \frac{1}{4}(\mathbf{e}_{24} + \mathbf{e}_{34} + \mathbf{e}_{124} + \mathbf{e}_{134})\}$.

2. *Two-sided ideal (division ring).* Find a list of two-sided ideal (division ring) K by multiplying the obtained left ideal elements by same idempotent from left, $K = P S = P\{Cl_{2,2}\}P$. The division ring of $Cl_{2,2}$ contains single element $K = \{K(1)\} = P_1 P = \frac{1}{4}(1 + \mathbf{e}_1 + \mathbf{e}_{23} + \mathbf{e}_{123})$. Being an idempotent, $K(1)$ represents a unit element of the real field, $K(1)^2 = K(1)$, which echoes the property $1 * 1 = 1$. Similarly, if K list would contain two elements (remind that elements that differ by sign are considered as same elements), then the other element will play the role of an imaginary unit. And, if K would contain four different elements it will be isomorphic to quaternion ring. The list of elements of the two sided ideal should be sorted according to their interpretation. In particular, the element that plays the role of real unit always appears first in the list. It is followed by elements that play the role of imaginary units. If K has four elements, the last three positions are occupied by non-commutative elements and (after proper reordering an sign change, when needed) can be interpreted as quaternion units q_1, q_2 and q_3 .

3. *Ideal basis.* The left ideal basis is obtained by comparing all the elements of ideal (starting from the first) and consequently dropping out all elements that can be obtained from previous elements by multiplying them by any division ring element. Since element choice may depend on its position in the ideal list, the element ordering plays an important role. It is convenient to sort the elements of ideal basis by ‘RevLex’ order what ensures that in spinor matrix representation all columns will be zero except the first one. Other orderings in ideal basis will give different matrix representations of the spinor. In $Cl_{2,2}$ case the division ring contains a single element, therefore, the ideal basis coincides with the ideal itself.
4. *Matrix rep of GA basis vectors.* Once the four-component ideal basis $S = \{S(1), S(2), S(3), S(4)\}$ is determined one can compute a matrix rep of basis vectors in algebraic spinor basis. The matrix entry (ij) can be calculated by $E_{ij}(\mathbf{e}_k) = \tilde{S}(i)^\# \mathbf{e}_k S(j)$, where $S(i)^\#$ is the ideal element in reciprocal basis. For example, if $S(1) = \frac{1}{4}(1 + \mathbf{e}_1 + \mathbf{e}_{23} + \mathbf{e}_{123})$ then $S(1)^\# = \frac{1}{4}(1 + \mathbf{e}^1 + \mathbf{e}^{23} + \mathbf{e}^{123}) = \frac{1}{4}(1 + \mathbf{e}_1 - \mathbf{e}_{23} - \mathbf{e}_{123})$, because in $Cl_{2,2}$ the orthonormal of reciprocal generators (basis) are $\mathbf{e}^1 = \mathbf{e}_1$, $\mathbf{e}^2 = \mathbf{e}_2$, $\mathbf{e}^3 = -\mathbf{e}_3$, and $\mathbf{e}^4 = -\mathbf{e}_4$. For each pair of indices (ij) in the ideal basis we obtain matrix entries $E_{ij}(\mathbf{e}_k)$ at fixed basis vector $\mathbf{e}_k$. In our case $i, j = 1, 2, 3, 4$ and $k = 1, 2, 3, 4$. The basis matrices $\hat{e}_1, \hat{e}_2, \hat{e}_3$, and $\hat{e}_4$ calculated with $E_{ij}(\mathbf{e}_k)$ then are, respectively,

$$\hat{e}_1 = \begin{bmatrix} K & 0 & 0 & 0 \\ 0 & -K & 0 & 0 \\ 0 & 0 & -K & 0 \\ 0 & 0 & 0 & K \end{bmatrix}, \quad \hat{e}_2 = \begin{bmatrix} 0 & K & 0 & 0 \\ K & 0 & 0 & 0 \\ 0 & 0 & 0 & K \\ 0 & 0 & K & 0 \end{bmatrix}, \quad \hat{e}_3 = \begin{bmatrix} 0 & -K & 0 & 0 \\ K & 0 & 0 & 0 \\ 0 & 0 & 0 & -K \\ 0 & 0 & K & 0 \end{bmatrix}, \quad \hat{e}_4 = \begin{bmatrix} 0 & 0 & -K & 0 \\ 0 & 0 & 0 & K \\ 0 & K & 0 & 0 \\ 0 & -K & 0 & 0 \end{bmatrix},$$

where K is a shortcut for division ring element $K = \{K(1)\}$, which in for $Cl_{2,2}$ is represented by a single entry 1 in 4×4 matrix E_{ij} . Thus, the matrix rep of GA basis elements (generators) in $Cl_{2,2}$ is

$$\hat{e}_1 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad \hat{e}_2 = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}, \quad \hat{e}_3 = \begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \end{bmatrix}, \quad \hat{e}_4 = \begin{bmatrix} 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{bmatrix}.$$

If the division ring K contains more elements, then all such elements will appear in the calculated $\hat{e}_i$ matrices. In order to get $\mathbb{R}$, $\mathbb{C}$, or $\mathbb{H}$ matrix reps we need to replace the division ring element by the corresponding isomorphic element (1) or (1, i), or $(q_0 = 1, q_1, q_2, q_3)$, where q_i denotes the quaternion components.

If $\mathbf{e}_k$ is replaced by elements of ideal basis S_k , namely $E_{ij}(S_k) = \tilde{S}^i S_k S_j$, then one can obtain the matrices having only a single unity in the first column,

$$E_{11} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad E_{21} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad E_{31} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad E_{41} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix},$$

what indicates that the ideal basis elements make up a spinorial basis.

If the division ring contains more than one element, i.e., it is a complex field or a quaternion ring, then the column spinor can be obtained by sum $\Psi = \sum_{j,k,i} s_{jk} K_i(k) S_1(j)$, where s_{jk} denotes an arbitrary real scalar and, K_i and S_i is the division ring and the ideal basis of the idempotent P_i (see the Spinor data tables for explicit P_i expression), respectively.

5. *General spinor Ψ .* A general spinor is the sum of generators (item 3) multiplied by real coefficients s_i ,

$$\begin{aligned} \Psi = & (s_1 + s_2 \mathbf{e}_2 + s_3 \mathbf{e}_4 + s_4 \mathbf{e}_{24}) P_1 = \\ & \frac{1}{4} s_1 (1 + \mathbf{e}_1 + \mathbf{e}_{23} + \mathbf{e}_{123}) + \frac{1}{4} s_2 (\mathbf{e}_2 + \mathbf{e}_3 - \mathbf{e}_{12} - \mathbf{e}_{13}) + \\ & \frac{1}{4} s_3 (\mathbf{e}_4 - \mathbf{e}_{14} + \mathbf{e}_{234} - \mathbf{e}_{1234}) + \frac{1}{4} s_4 (\mathbf{e}_{24} + \mathbf{e}_{34} + \mathbf{e}_{124} + \mathbf{e}_{134}). \end{aligned}$$

6. *Spinor Ψ in matrix form.* In the matrix form the general MV spinor can be expanded in the ideal basis as

$$\hat{\Psi} = s_1 E_{11} + s_2 E_{21} + s_3 E_{31} + s_4 E_{41}.$$

7. *Square of spinor norm* is

$$\|\Psi\|^2 = \langle \Psi^\dagger \Psi \rangle = \langle \tilde{\Psi}^\# \Psi \rangle = -\langle \mathbf{e}_{34} \tilde{\Psi} \mathbf{e}_{34} \Psi \rangle = \frac{1}{4} (s_1^2 + s_2^2 + s_3^2 + s_4^2),$$

where the tilde and brackets indicate, respectively, the reversion and scalar part projection.

The results of the example are summarized in Table VI ($Cl_{2,2}$ algebra). The implementation and step by step computations are given in⁴ (the notebook *10_AlgebraicSpinorsOfRealCl.nb*), where data for GAs up to $n = 8$ are explicitly computed and presented.

B. Norm of spinor in real GA.

In spinor data Tables the MVs have standard form, i.e., they are written with lower indices. The norm may be rewritten more uniformly in algebra independent way if upper (related to reciprocal basis) and lower indices are used simultaneously. Then, with the reciprocal orthonormalized frame introduced the spinor norm is $\|\Psi\|^2 = \langle \Psi^\dagger \Psi \rangle = \langle \widetilde{\Psi}^\sharp \Psi \rangle_0$, where Ψ is a general spinor and $\Psi^\sharp$ is the spinor with all indices raised, i.e. in the reciprocal basis (cf. item 4 in the Examples IV A). Then, the spinor $\Psi^\sharp$ with all upper indices after index lowering gives a correct sign.

The normalization factor 2^k in the real spinor data Tables (last row, left-side of equation), where k counts a number of factors in the primitive idempotent P_i , ensures that the hermitian norm computed either by formula $\|\Psi\|^2 = \langle \Psi^\dagger \Psi \rangle = \langle \widetilde{\Psi}^\sharp \Psi \rangle$, or its matrix counterpart $\|\hat{\Psi}\|^2 = 2^{-k} \text{Tr}(\hat{\Psi}^\dagger \hat{\Psi})$ coincides. In real Euclidean spaces the norm is $\|\Psi\|^2 = \langle \widetilde{\Psi} \Psi \rangle$, since index raising/lowering operation has no effect upon sign of basis elements⁵.

V. COMPLEX GA SPINORS DATA TABLES (COMPLEX IDEMPOTENT)

In real GAs, not all basis elements are acceptable for construction of primitive idempotents. Specially, the squares of elements are required to be equal to +1. If multiplication of basis elements by imaginary unit is permitted, then any $\mathbf{e}_j$ may be used in idempotent construction. Complex spinor data Tables were constructed by allowing complex idempotents. Both the computation procedure and the notations in the Tables below remain the same. The asterisk (*) denotes complex conjugation.

A. Even complex algebras

	$Cl_{2,0}[\mathbb{C}(2)]$	$Cl_{1,1}[\mathbb{C}(2)]$	$Cl_{0,2}[\mathbb{C}(2)]$
1	$t = \frac{1}{2}(1 + \mathbf{e}_1)$	$t = \frac{1}{2}(1 + \mathbf{e}_1)$	$t = \frac{1}{2}(1 + i\mathbf{e}_1)$
2	$\{t1t\} = \frac{1}{2}(1 + i\mathbf{e}_1)$	$\{t1t\} = \frac{1}{2}(1 + \mathbf{e}_1)$	$\{t1t\} = \frac{1}{2}(1 + i\mathbf{e}_1)$
3	$\{t, \mathbf{e}_2t\}$	$\{t, \mathbf{e}_2t\}$	$\{t, \mathbf{e}_2t\}$
4	$\{E_{11} - E_{22}, E_{12} + E_{21}\}$	$\{E_{11} - E_{22}, -E_{12} + E_{21}\}$	$\{i(-E_{11} + E_{22}), -E_{12} + E_{21}\}$
5	$\Psi = (s_1 + s_2\mathbf{e}_2)t$	$\Psi = (s_1 + s_2\mathbf{e}_2)t$	$\Psi = (s_1 + s_2\mathbf{e}_2)t$
6	$s_1E_{11} + s_2E_{21}$	$s_1E_{11} + s_2E_{21}$	$s_1E_{11} + s_2E_{21}$
7	$\langle \widetilde{\Psi} \Psi \rangle = \frac{1}{2}(s_1 ^2 + s_2 ^2)$	$\langle \mathbf{e}_1 \widetilde{\Psi} \mathbf{e}_1 \Psi \rangle = \frac{1}{2}(s_1 ^2 + s_2 ^2)$	$\langle \widetilde{\Psi}^* \Psi \rangle = \frac{1}{2}(s_1 ^2 + s_2 ^2)$

TABLE X: Complex spinor tables for GAs with vector space dimension $n = 2$

$Cl_{4,0}[\mathbb{C}(4)]$	$Cl_{3,1}[\mathbb{C}(4)]$
1 $t = \frac{1}{4}(1 + \mathbf{e}_1)(1 + \mathbf{ie}_{23})$	$t = \frac{1}{4}(1 + \mathbf{e}_1)(1 + \mathbf{ie}_{23})$
2 $\{t1t\}$	$\{t1t\}$
3 $\{t, \mathbf{ie}_2t, \mathbf{ie}_4t, \mathbf{e}_{24}t\}$	$\{t, \mathbf{ie}_2t, \mathbf{ie}_4t, \mathbf{e}_{24}t\}$
4 $\{E_{11} - E_{22} - E_{33} + E_{44}, \mathbf{i}(E_{12} - E_{21} - E_{34} + E_{43}),$ $-E_{12} - E_{21} + E_{34} + E_{43}, \mathbf{i}(E_{13} + E_{24} - E_{31} - E_{42})\}$	$\{E_{11} - E_{22} - E_{33} + E_{44}, \mathbf{i}(E_{12} - E_{21} - E_{34} + E_{43}),$ $-E_{12} - E_{21} + E_{34} + E_{43}, -\mathbf{i}(E_{13} + E_{24} + E_{31} + E_{42})\}$
5 $\Psi = (s_1 + \mathbf{is}_2\mathbf{e}_2 + \mathbf{is}_3\mathbf{e}_4 + s_4\mathbf{e}_{24})t$	$\Psi = (s_1 + \mathbf{is}_2\mathbf{e}_2 + \mathbf{is}_3\mathbf{e}_4 + s_4\mathbf{e}_{24})t$
6 $s_1E_{11} + s_2E_{21} + s_3E_{31} + s_4E_{41}$	$s_1E_{11} + s_2E_{21} + s_3E_{31} + s_4E_{41}$
7 $\langle \widetilde{\Psi}^* \Psi \rangle = \frac{1}{4}(s_1 ^2 + \dots + s_4 ^2)$	$-\langle \mathbf{e}_4 \widetilde{\Psi}^* \mathbf{e}_4 \Psi \rangle = \frac{1}{4}(s_1 ^2 + \dots + s_4 ^2)$
$Cl_{2,2}[\mathbb{C}(4)]$	$Cl_{1,3}[\mathbb{C}(4)]$
1 $t = \frac{1}{4}(1 + \mathbf{e}_1)(1 + \mathbf{e}_{23})$	$t = \frac{1}{4}(1 + \mathbf{e}_1)(1 + \mathbf{ie}_{23})$
2 $\{t1t\}$	$\{t1t\}$
3 $\{t, \mathbf{e}_2t, \mathbf{e}_4t, \mathbf{e}_{24}t\}$	$\{t, \mathbf{ie}_2t, \mathbf{ie}_4t, \mathbf{e}_{24}t\}$
4 $\{E_{11} - E_{22} - E_{33} + E_{44}, E_{12} + E_{21} + E_{34} + E_{43},$ $-E_{12} + E_{21} - E_{34} + E_{43}, -E_{13} + E_{24} + E_{31} - E_{42}\}$	$\{E_{11} - E_{22} - E_{33} + E_{44}, \mathbf{i}(-E_{12} - E_{21} + E_{34} + E_{43}),$ $-E_{12} + E_{21} + E_{34} - E_{43}, -\mathbf{i}(E_{13} + E_{24} + E_{31} + E_{42})\}$
5 $\Psi = (s_1 + s_2\mathbf{e}_2 + s_3\mathbf{e}_4 + s_4\mathbf{e}_{24})t$	$\Psi = (s_1 + \mathbf{is}_2\mathbf{e}_2 + \mathbf{is}_3\mathbf{e}_4 + s_4\mathbf{e}_{24})t$
6 $s_1E_{11} + s_2E_{21} + s_3E_{31} + s_4E_{41}$	$s_1E_{11} + s_2E_{21} + s_3E_{31} + s_4E_{41}$
7 $-\langle \mathbf{e}_{34} \widetilde{\Psi}^* \mathbf{e}_{34} \Psi \rangle = \frac{1}{4}(s_1 ^2 + \dots + s_4 ^2)$	$\langle \mathbf{e}_1 \widetilde{\Psi}^* \mathbf{e}_1 \Psi \rangle = \frac{1}{4}(s_1 ^2 + \dots + s_4 ^2)$
$Cl_{0,4}[\mathbb{C}(4)]$	
1 $t = \frac{1}{4}(1 + \mathbf{ie}_1)(1 + \mathbf{ie}_{23})$	
2 $\{t1t\}$	
3 $\{t, \mathbf{e}_2t, \mathbf{ie}_4t, \mathbf{ie}_{24}t\}$	
4 $\{\mathbf{i}(-E_{11} + E_{22} + E_{33} - E_{44}), -E_{12} + E_{21} - E_{34} + E_{43}, \mathbf{i}(E_{12} + E_{21} + E_{34} + E_{43}), \mathbf{i}(-E_{13} + E_{24} - E_{31} + E_{42})\}$	
5 $\Psi = (s_1 + s_2\mathbf{e}_2 + \mathbf{is}_3\mathbf{e}_4 + \mathbf{is}_4\mathbf{e}_{24})t$	
6 $s_1E_{11} + s_2E_{21} + s_3E_{31} + s_4E_{41}$	
7 $\langle \widetilde{\Psi}^* \Psi \rangle = \frac{1}{4}(s_1 ^2 + \dots + s_4 ^2)$	

TABLE XI: Complex spinor tables for GAs with vector space dimension $n = 4$

$Cl_{6,0}[\mathbb{C}(8)]$	
1	$t = \frac{1}{8}(1 + \mathbf{e}_1)(1 + i\mathbf{e}_{23})(1 + i\mathbf{e}_{45})$
2	$\{t1t\}$
3	$\{t, \mathbf{e}_2t, \mathbf{e}_4t, \mathbf{e}_{24}t, \mathbf{e}_6t, \mathbf{e}_{26}t, \mathbf{e}_{46}t, \mathbf{e}_{246}t\}$
4	$\{E_{11} - E_{22} - E_{33} + E_{44} - E_{55} + E_{66} + E_{77} - E_{88}, E_{12} + E_{21} + E_{34} + E_{43} + E_{56} + E_{65} + E_{78} + E_{87},$ $i(E_{12} - E_{21} + E_{34} - E_{43} + E_{56} - E_{65} + E_{78} - E_{87}), E_{13} - E_{24} + E_{31} - E_{42} + E_{57} - E_{68} + E_{75} - E_{86},$ $i(E_{13} - E_{24} - E_{31} + E_{42} + E_{57} - E_{68} - E_{75} + E_{86}), E_{15} - E_{26} - E_{37} + E_{48} + E_{51} - E_{62} - E_{73} + E_{84}\}$
5	$\Psi = (s_1 + s_2\mathbf{e}_2 + s_3\mathbf{e}_4 + s_4\mathbf{e}_{24} + s_5\mathbf{e}_6 + s_6\mathbf{e}_{26} + s_7\mathbf{e}_{46} + s_8\mathbf{e}_{246})t$
6	$s_1E_{11} + s_2E_{21} + s_3E_{31} + s_4E_{41} + s_5E_{51} + s_6E_{61} + s_7E_{71} + s_8E_{81}$
7	$\langle \widetilde{\Psi}^* \Psi \rangle = \frac{1}{8}(s_1 ^2 + \dots + s_8 ^2)$
$Cl_{5,1}[\mathbb{C}(8)]$	
1	$t = \frac{1}{8}(1 + \mathbf{e}_1)(1 + i\mathbf{e}_{23})(1 + i\mathbf{e}_{45})$
2	$\{t1t\}$
3	$\{t, \mathbf{e}_2t, \mathbf{e}_4t, \mathbf{e}_{24}t, \mathbf{e}_6t, \mathbf{e}_{26}t, \mathbf{e}_{46}t, \mathbf{e}_{246}t\}$
4	$\{E_{11} - E_{22} - E_{33} + E_{44} - E_{55} + E_{66} + E_{77} - E_{88}, E_{12} + E_{21} + E_{34} + E_{43} + E_{56} + E_{65} + E_{78} + E_{87},$ $i(E_{12} - E_{21} + E_{34} - E_{43} + E_{56} - E_{65} + E_{78} - E_{87}), E_{13} - E_{24} + E_{31} - E_{42} + E_{57} - E_{68} + E_{75} - E_{86},$ $i(E_{13} - E_{24} - E_{31} + E_{42} + E_{57} - E_{68} - E_{75} + E_{86}), -E_{15} + E_{26} + E_{37} - E_{48} + E_{51} - E_{62} - E_{73} + E_{84}\}$
5	$\Psi = (s_1 + s_2\mathbf{e}_2 + s_3\mathbf{e}_4 + s_4\mathbf{e}_{24} + s_5\mathbf{e}_6 + s_6\mathbf{e}_{26} + s_7\mathbf{e}_{46} + s_8\mathbf{e}_{246})t$
6	$s_1E_{11} + s_2E_{21} + s_3E_{31} + s_4E_{41} + s_5E_{51} + s_6E_{61} + s_7E_{71} + s_8E_{81}$
7	$-\langle \widetilde{\Psi}^* \mathbf{e}_6 \Psi \rangle = \frac{1}{8}(s_1 ^2 + \dots + s_8 ^2)$
$Cl_{4,2}[\mathbb{C}(8)]$	
1	$t = \frac{1}{8}(1 + \mathbf{e}_1)(1 + i\mathbf{e}_{23})(1 + \mathbf{e}_{45})$
2	$\{t1t\}$
3	$\{t, i\mathbf{e}_2t, i\mathbf{e}_4t, i\mathbf{e}_{24}t, i\mathbf{e}_6t, i\mathbf{e}_{26}t, i\mathbf{e}_{46}t, i\mathbf{e}_{246}t\}$
4	$\{E_{11} - E_{22} - E_{33} + E_{44} - E_{55} + E_{66} + E_{77} - E_{88}, i(E_{12} - E_{21} - E_{34} + E_{43} - E_{56} + E_{65} + E_{78} - E_{87}),$ $-E_{12} - E_{21} + E_{34} + E_{43} + E_{56} + E_{65} - E_{78} - E_{87}, i(E_{13} + E_{24} - E_{31} - E_{42} - E_{57} - E_{68} + E_{75} + E_{86}),$ $i(-E_{13} - E_{24} - E_{31} - E_{42} + E_{57} + E_{68} - E_{75} + E_{86}), i(-E_{15} - E_{26} - E_{37} - E_{48} - E_{51} - E_{62} - E_{73} - E_{84})\}$
5	$\Psi = (s_1 + i s_2\mathbf{e}_2 + i s_3\mathbf{e}_4 + s_4\mathbf{e}_{24} + i s_5\mathbf{e}_6 + s_6\mathbf{e}_{26} + s_7\mathbf{e}_{46} + i s_8\mathbf{e}_{246})t$
6	$s_1E_{11} + s_2E_{21} + s_3E_{31} + s_4E_{41} + s_5E_{51} + s_6E_{61} + s_7E_{71} + s_8E_{81}$
7	$-\langle \mathbf{e}_{56} \widetilde{\Psi}^* \mathbf{e}_{56} \Psi \rangle = \frac{1}{8}(s_1 ^2 + \dots + s_8 ^2)$
$Cl_{3,3}[\mathbb{C}(8)]$	
1	$t = \frac{1}{8}(1 + \mathbf{e}_1)(1 + i\mathbf{e}_{23})(1 + i\mathbf{e}_{45})$
2	$\{t1t\}$
3	$\{t, \mathbf{e}_2t, \mathbf{e}_4t, i\mathbf{e}_{24}t, \mathbf{e}_6t, \mathbf{e}_{26}t, i\mathbf{e}_{46}t, \mathbf{e}_{246}t\}$
4	$\{E_{11} - E_{22} - E_{33} + E_{44} - E_{55} + E_{66} + E_{77} - E_{88}, E_{12} + E_{21} + iE_{34} - iE_{43} + E_{56} + E_{65} - iE_{78} + iE_{87},$ $iE_{12} - iE_{21} - E_{34} + E_{43} + iE_{56} - iE_{65} + E_{78} + E_{87}, -E_{13} + iE_{24} + E_{31} + iE_{42} - iE_{57} + E_{68} - iE_{75} - E_{86},$ $iE_{13} + E_{24} + iE_{31} - E_{42} - E_{57} - iE_{68} + E_{75} - iE_{86}, -E_{15} + E_{26} + iE_{37} + iE_{48} + E_{51} - E_{62} + iE_{73} + iE_{84}\}$
5	$\Psi = (s_1 + s_2\mathbf{e}_2 + s_3\mathbf{e}_4 + i s_4\mathbf{e}_{24} + s_5\mathbf{e}_6 + s_6\mathbf{e}_{26} + i s_7\mathbf{e}_{46} + s_8\mathbf{e}_{246})t$
6	$s_1E_{11} + s_2E_{21} + s_3E_{31} + s_4E_{41} + s_5E_{51} + s_6E_{61} + s_7E_{71} + s_8E_{81}$
7	$-\langle \mathbf{e}_{123} \widetilde{\Psi}^* \mathbf{e}_{123} \Psi \rangle = \frac{1}{8}(s_1 ^2 + \dots + s_8 ^2)$
$Cl_{2,4}[\mathbb{C}(8)]$	
1	$t = \frac{1}{8}(1 + \mathbf{e}_1)(1 + \mathbf{e}_{23})(1 + i\mathbf{e}_{45})$
2	$\{t1t\}$
3	$\{t, i\mathbf{e}_2t, i\mathbf{e}_4t, \mathbf{e}_{24}t, i\mathbf{e}_6t, \mathbf{e}_{26}t, \mathbf{e}_{46}t, i\mathbf{e}_{246}t\}$
4	$\{E_{11} - E_{22} - E_{33} + E_{44} - E_{55} + E_{66} + E_{77} - E_{88}, i(E_{12} - E_{21} - E_{34} + E_{43} - E_{56} + E_{65} + E_{78} - E_{87}),$ $i(-E_{12} - E_{21} + E_{34} + E_{43} + E_{56} + E_{65} - E_{78} - E_{87}), i(-E_{13} - E_{24} - E_{31} - E_{42} + E_{57} + E_{68} + E_{75} + E_{86}),$ $-E_{13} - E_{24} + E_{31} + E_{42} + E_{57} + E_{68} - E_{75} - E_{86}, i(-E_{15} - E_{26} - E_{37} - E_{48} - E_{51} - E_{62} - E_{73} - E_{84})\}$
5	$\Psi = (s_1 + i s_2\mathbf{e}_2 + i s_3\mathbf{e}_4 + s_4\mathbf{e}_{24} + i s_5\mathbf{e}_6 + s_6\mathbf{e}_{26} + s_7\mathbf{e}_{46} + i s_8\mathbf{e}_{246})t$
6	$s_1E_{11} + s_2E_{21} + s_3E_{31} + s_4E_{41} + s_5E_{51} + s_6E_{61} + s_7E_{71} + s_8E_{81}$
7	$-\langle \mathbf{e}_{12} \widetilde{\Psi}^* \mathbf{e}_{12} \Psi \rangle = \frac{1}{8}(s_1 ^2 + \dots + s_8 ^2)$
$Cl_{1,5}[\mathbb{C}(8)]$	
1	$t = \frac{1}{8}(1 + \mathbf{e}_1)(1 + i\mathbf{e}_{23})(1 + i\mathbf{e}_{45})$
2	$\{t1t\}$
3	$\{t, \mathbf{e}_2t, \mathbf{e}_4t, \mathbf{e}_{24}t, \mathbf{e}_6t, i\mathbf{e}_{26}t, i\mathbf{e}_{46}t, \mathbf{e}_{246}t\}$
4	$\{E_{11} - E_{22} - E_{33} + E_{44} - E_{55} + E_{66} + E_{77} - E_{88}, -E_{12} + E_{21} - E_{34} + E_{43} - iE_{56} - iE_{65} + iE_{78} + iE_{87},$ $iE_{12} + iE_{21} + iE_{34} + iE_{43} - E_{56} + E_{65} + E_{78} - E_{87}, -E_{13} + E_{24} + E_{31} - E_{42} - iE_{57} - iE_{68} - iE_{75} - iE_{86},$ $iE_{13} - iE_{24} + iE_{31} - iE_{42} - E_{57} - E_{68} + E_{75} + E_{86}, -E_{15} + iE_{26} + iE_{37} - E_{48} + E_{51} + iE_{62} + iE_{73} + E_{84}\}$
5	$\Psi = (s_1 + s_2\mathbf{e}_2 + s_3\mathbf{e}_4 + s_4\mathbf{e}_{24} + s_5\mathbf{e}_6 + i s_6\mathbf{e}_{26} + i s_7\mathbf{e}_{46} + s_8\mathbf{e}_{246})t$
6	$s_1E_{11} + s_2E_{21} + s_3E_{31} + s_4E_{41} + s_5E_{51} + s_6E_{61} + s_7E_{71} + s_8E_{81}$
7	$\langle \mathbf{e}_1 \widetilde{\Psi}^* \mathbf{e}_1 \Psi \rangle = \frac{1}{8}(s_1 ^2 + \dots + s_8 ^2)$

TABLE XII: Complex spinor tables for $Cl_{6,0}$, $Cl_{5,1}$, $Cl_{4,2}$, $Cl_{3,3}$, $Cl_{2,4}$ and $Cl_{1,5}$; $n = 6$

	$Cl_{0,6}[\mathbb{C}(8)]$
1	$t = \frac{1}{8}(1 + \mathbf{ie}_1)(1 + \mathbf{ie}_{23})(1 + \mathbf{ie}_{45})$
2	$\{t1t\}$
3	$\{it, \mathbf{ie}_2t, \mathbf{ie}_4t, \mathbf{ie}_{24}t, \mathbf{ie}_6t, \mathbf{ie}_{26}t, \mathbf{ie}_{46}t, \mathbf{ie}_{246}t\}$
4	$\{i(-E_{11} + E_{22} + E_{33} - E_{44} + E_{55} - E_{66} - E_{77} + E_{88}), -E_{12} + E_{21} - E_{34} + E_{43} - E_{56} + E_{65} - E_{78} + E_{87},$ $i(E_{12} + E_{21} + E_{34} + E_{43} + E_{56} + E_{65} + E_{78} + E_{87}), -E_{13} + E_{24} + E_{31} - E_{42} - E_{57} + E_{68} + E_{75} - E_{86},$ $i(E_{13} - E_{24} + E_{31} - E_{42} + E_{57} - E_{68} + E_{75} - E_{86}), -E_{15} + E_{26} + E_{37} - E_{48} + E_{51} - E_{62} - E_{73} + E_{84}\}$
5	$\Psi = i(s_1 + s_2\mathbf{e}_2 + s_3\mathbf{e}_4 + s_4\mathbf{e}_{24} + s_5\mathbf{e}_6 + s_6\mathbf{e}_{26} + s_7\mathbf{e}_{46} + s_8\mathbf{e}_{246})t$
6	$i(s_1E_{11} + s_2E_{21} + s_3E_{31} + s_4E_{41} + s_5E_{51} + s_6E_{61} + s_7E_{71} + s_8E_{81})$
7	$\langle \widetilde{\Psi}^* \Psi \rangle = \frac{1}{8}(s_1 ^2 + \dots + s_8 ^2)$

TABLE XIII: Complex spinor tables for $Cl_{0,6}$; $n=6$

B. Odd complex algebras

In matrix rep, the odd complex algebras consist of two blocks on a diagonal. The diagonal elements in the first and second block come from, respectively, left basis ideal sorted in ‘RevLex’ order and after application of grade inversion to left ideal basis, which then is sorted in ‘InvLex’ order. Such a procedure ensures that in spinor matrix rep the first nonzero column of spinor always appears as a first column, and the second nonzero column appears in the middle of the matrix, i.e., the spinor matches the block diagonal structure of MV matrix rep). The coefficients s_i now are complex numbers. The block diagonal structure reflects the fact that the underlying field $\{1, i\}$ is commutative. Contrary to the case of real algebras, the block diagonal matrices, $\mathbb{C}(n) \oplus \mathbb{C}(n)$, of the complex spinor have n independent coefficients instead of $2n$. Let us remind that in a semi-simple real algebra (i.e. when $p - q = 1 \bmod 4$), the spinors have $2n$ free coefficients.

	$Cl_{3,0}[\mathbb{C}(2) \oplus \mathbb{C}(2)]$		$Cl_{2,1}[\mathbb{C}(2) \oplus \mathbb{C}(2)]$
1	$t = \frac{1}{2}(1 + \mathbf{e}_1)$		$t = \frac{1}{2}(1 + \mathbf{e}_1)$
2	$\{t1t, t\mathbf{e}_{23}t\}$		$\{t1t, t\mathbf{ie}_{23}t\}$
3	$\{t, \mathbf{e}_2t\}$		$\{t, \mathbf{e}_2t\}$
4	$\{E_{11} - E_{22} + E_{33} - E_{44}, E_{12} + E_{21} - E_{34} - E_{43},$ $i(-E_{12} + E_{21} - E_{34} + E_{43})\}$		$\{E_{11} - E_{22} + E_{33} - E_{44}, E_{12} + E_{21} - E_{34} - E_{43},$ $-E_{12} + E_{21} - E_{34} + E_{43}\}$
5	$\Psi = (s_1 + s_2\mathbf{e}_{23} + s_3\mathbf{e}_2 - s_4\mathbf{e}_3)t$		$\Psi = (s_1 + is_2\mathbf{e}_{23} + s_3\mathbf{e}_2 - is_4\mathbf{e}_3)t$
6	$(s_1 + is_2)E_{11} + (s_3 - is_4)E_{21}$ $+ (s_1 - is_2)E_{33} + (-s_3 - is_4)E_{43}$		$(s_1 + is_2)E_{11} + (s_3 - is_4)E_{21}$ $+ (s_1 - is_2)E_{33} + (-s_3 - is_4)E_{43}$
7	$\langle \widetilde{\Psi}^* \Psi \rangle = \frac{1}{2}(s_1 ^2 + \dots + s_4 ^2)$		$\langle \widetilde{\Psi}^* \Psi \rangle = \frac{1}{2}(s_1 ^2 + \dots + s_4 ^2)$

	$Cl_{1,2}[\mathbb{C}(2) \oplus \mathbb{C}(2)]$		$Cl_{0,3}[\mathbb{C}(2) \oplus \mathbb{C}(2)]$
1	$t = \frac{1}{2}(1 + \mathbf{e}_1)$		$t = \frac{1}{2}(1 + \mathbf{ie}_1)$
2	$\{t1t, t\mathbf{e}_{23}t\}$		$\{t1t, t\mathbf{e}_{23}t\}$
3	$\{t, \mathbf{e}_2t\}$		$\{t, \mathbf{e}_2t\}$
4	$\{E_{11} - E_{22} + E_{33} - E_{44}, -E_{12} + E_{21} - E_{34} + E_{43},$ $i(-E_{12} - E_{21} + E_{34} + E_{43})\}$		$\{i(-E_{11} + E_{22} - E_{33} + E_{44}), -E_{12} + E_{21} - E_{34} + E_{43},$ $i(-E_{12} - E_{21} + E_{34} + E_{43})\}$
5	$\Psi = (s_1 + s_2\mathbf{e}_{23} + s_3\mathbf{e}_2 + s_4\mathbf{e}_3)t$		$\Psi = (s_1 + s_2\mathbf{e}_{23} + s_3\mathbf{e}_2 + s_4\mathbf{e}_3)t$
6	$(s_1 + is_2)E_{11} + (s_3 - is_4)E_{21}$ $+ (s_1 - is_2)E_{33} + (s_3 + is_4)E_{43}$		$(s_1 + is_2)E_{11} + (s_3 - is_4)E_{21}$ $+ (s_1 - is_2)E_{33} + (s_3 + is_4)E_{43}$
7	$\langle \mathbf{e}_1 \widetilde{\Psi}^* \mathbf{e}_1 \Psi \rangle = \frac{1}{2}(s_1 ^2 + \dots + s_4 ^2)$		$\langle \widetilde{\Psi}^* \Psi \rangle = \frac{1}{2}(s_1 ^2 + \dots + s_4 ^2)$

TABLE XIV: Complex spinor tables for geometric algebras with vector space dimension $n = 3$

$Cl_{5,0}[\mathbb{C}(4) \oplus \mathbb{C}(4)]$	$Cl_{4,1}[\mathbb{C}(4) \oplus \mathbb{C}(4)]$
1 $t = \frac{1}{4}(1 + \mathbf{e}_1)(1 + \mathbf{ie}_{23})$	$t = \frac{1}{4}(1 + \mathbf{e}_1)(1 + \mathbf{ie}_{23})$
2 $\{t1t, t\mathbf{e}_{45}t\}$	$\{t1t, t(\mathbf{ie}_{45})t\}$
3 $\{t, \mathbf{ie}_2t, \mathbf{ie}_4t, \mathbf{e}_{24}t\}$	$\{t, \mathbf{ie}_2t, \mathbf{ie}_4t, \mathbf{e}_{24}t\}$
4 $\{E_{11} - E_{22} - E_{33} + E_{44} + E_{55} - E_{66} - E_{77} + E_{88},$ $i(E_{12} - E_{21} - E_{34} + E_{43} + E_{56} - E_{65} - E_{78} + E_{87}),$ $-E_{12} - E_{21} + E_{34} + E_{43} - E_{56} - E_{65} + E_{78} + E_{87},$ $i(E_{13} + E_{24} - E_{31} - E_{42} + E_{57} + E_{68} - E_{75} - E_{86}),$ $E_{13} + E_{24} + E_{31} + E_{42} - E_{57} - E_{68} - E_{75} - E_{86}\}$	$\{E_{11} - E_{22} - E_{33} + E_{44} + E_{55} - E_{66} - E_{77} + E_{88},$ $i(E_{12} - E_{21} - E_{34} + E_{43} + E_{56} - E_{65} - E_{78} + E_{87}),$ $-E_{12} - E_{21} + E_{34} + E_{43} - E_{56} - E_{65} + E_{78} + E_{87},$ $i(E_{13} + E_{24} - E_{31} - E_{42} + E_{57} + E_{68} - E_{75} - E_{86}),$ $i(-E_{13} - E_{24} - E_{31} - E_{42} + E_{57} + E_{68} + E_{75} + E_{86})\}$
5 $\Psi = (s_1 + s_2\mathbf{e}_{45} + \mathbf{is}_3\mathbf{e}_2 + \mathbf{is}_4\mathbf{e}_{245} + \mathbf{is}_5\mathbf{e}_4 - \mathbf{is}_6\mathbf{e}_5$ $+ s_7\mathbf{e}_{24} - s_8\mathbf{e}_{25})t$	$\Psi = (s_1 + \mathbf{is}_2\mathbf{e}_{45} + \mathbf{is}_3\mathbf{e}_2 - s_4\mathbf{e}_{245} + \mathbf{is}_5\mathbf{e}_4 + s_6\mathbf{e}_5$ $+ s_7\mathbf{e}_{24} - \mathbf{is}_8\mathbf{e}_{25})t$
6 $(s_1 + \mathbf{is}_2)E_{11} + (s_3 + \mathbf{is}_4)E_{21} + (s_5 - \mathbf{is}_6)E_{31}$ $+ (s_7 - \mathbf{is}_8)E_{41} + (s_1 - \mathbf{is}_2)E_{55} + (s_3 - \mathbf{is}_4)E_{65}$ $+ (s_5 + \mathbf{is}_6)E_{75} + (s_7 + \mathbf{is}_8)E_{85}$	$(s_1 + \mathbf{is}_2)E_{11} + (s_3 + \mathbf{is}_4)E_{21} + (s_5 - \mathbf{is}_6)E_{31}$ $+ (s_7 - \mathbf{is}_8)E_{41} + (s_1 - \mathbf{is}_2)E_{55} + (s_3 - \mathbf{is}_4)E_{65}$ $+ (s_5 + \mathbf{is}_6)E_{75} + (s_7 + \mathbf{is}_8)E_{85}$
7 $\langle \widetilde{\Psi}^* \Psi \rangle = \frac{1}{4}(s_1 ^2 + \dots + s_8 ^2)$	$-\langle \widetilde{\mathbf{e}_5} \widetilde{\Psi}^* \mathbf{e}_5 \Psi \rangle = \frac{1}{4}(s_1 ^2 + \dots + s_8 ^2)$
$Cl_{3,2}[\mathbb{C}(4) \oplus \mathbb{C}(4)]$	$Cl_{2,3}[\mathbb{C}(4) \oplus \mathbb{C}(4)]$
1 $t = \frac{1}{4}(1 + \mathbf{e}_1)(1 + \mathbf{ie}_{23})$	$t = \frac{1}{4}(1 + \mathbf{e}_1)(1 + \mathbf{e}_{23})$
2 $\{t1t, t\mathbf{e}_{45}t\}$	$\{t1t, t\mathbf{e}_{45}t\}$
3 $\{t, \mathbf{ie}_2t, \mathbf{ie}_4t, \mathbf{e}_{24}t\}$	$\{t, \mathbf{e}_2t, \mathbf{e}_4t, \mathbf{e}_{24}t\}$
4 $\{E_{11} - E_{22} - E_{33} + E_{44} + E_{55} - E_{66} - E_{77} + E_{88},$ $i(E_{12} - E_{21} - E_{34} + E_{43} + E_{56} - E_{65} - E_{78} + E_{87}),$ $-E_{12} - E_{21} + E_{34} + E_{43} - E_{56} - E_{65} + E_{78} + E_{87},$ $i(-E_{13} - E_{24} - E_{31} - E_{42} + E_{57} + E_{68} + E_{75} - E_{86}),$ $E_{13} + E_{24} - E_{31} - E_{42} + E_{57} + E_{68} - E_{75} - E_{86}\}$	$\{E_{11} - E_{22} - E_{33} + E_{44} + E_{55} - E_{66} - E_{77} + E_{88},$ $E_{12} + E_{21} + E_{34} + E_{43} - E_{56} - E_{65} - E_{78} - E_{87},$ $-E_{12} + E_{21} - E_{34} + E_{43} + E_{56} - E_{65} + E_{78} - E_{87},$ $-E_{13} + E_{24} + E_{31} - E_{42} - E_{57} + E_{68} + E_{75} - E_{86},$ $i(-E_{13} + E_{24} - E_{31} + E_{42} + E_{57} - E_{68} + E_{75} - E_{86})\}$
5 $\Psi = (s_1 + s_2\mathbf{e}_{45} + \mathbf{is}_3\mathbf{e}_2 + \mathbf{is}_4\mathbf{e}_{245} + \mathbf{is}_5\mathbf{e}_4 + \mathbf{is}_6\mathbf{e}_5$ $+ s_7\mathbf{e}_{24} + s_8\mathbf{e}_{25})t$	$\Psi = (s_1 + s_2\mathbf{e}_{45} + s_3\mathbf{e}_2 + s_4\mathbf{e}_{245} + s_5\mathbf{e}_4 + s_6\mathbf{e}_5$ $+ s_7\mathbf{e}_{24} + s_8\mathbf{e}_{25})t$
6 $(s_1 + \mathbf{is}_2)E_{11} + (s_3 + \mathbf{is}_4)E_{21} + (s_5 - \mathbf{is}_6)E_{31}$ $+ (s_7 - \mathbf{is}_8)E_{41} + (s_1 - \mathbf{is}_2)E_{55} + (s_3 - \mathbf{is}_4)E_{65}$ $+ (-s_5 - \mathbf{is}_6)E_{75} + (-s_7 - \mathbf{is}_8)E_{85}$	$(s_1 + \mathbf{is}_2)E_{11} + (s_3 + \mathbf{is}_4)E_{21} + (s_5 - \mathbf{is}_6)E_{31}$ $+ (s_7 - \mathbf{is}_8)E_{41} + (s_1 - \mathbf{is}_2)E_{55} + (-s_3 + \mathbf{is}_4)E_{65}$ $+ (s_5 + \mathbf{is}_6)E_{75} + (-s_7 - \mathbf{is}_8)E_{85}$
7 $-\langle \mathbf{e}_{45} \widetilde{\Psi}^* \mathbf{e}_{45} \Psi \rangle = \frac{1}{4}(s_1 ^2 + \dots + s_8 ^2)$	$-\langle \mathbf{e}_{12} \widetilde{\Psi}^* \mathbf{e}_{12} \Psi \rangle = \frac{1}{4}(s_1 ^2 + \dots + s_8 ^2)$
$Cl_{1,4}[\mathbb{C}(4) \oplus \mathbb{C}(4)]$	$Cl_{0,5}[\mathbb{C}(4) \oplus \mathbb{C}(4)]$
1 $t = \frac{1}{4}(1 + \mathbf{e}_1)(1 + \mathbf{ie}_{23})$	$t = \frac{1}{4}(1 + \mathbf{ie}_1)(1 + \mathbf{ie}_{23})$
2 $\{t1t, t\mathbf{e}_{45}t\}$	$\{t1t, t\mathbf{e}_{45}t\}$
3 $\{t, \mathbf{ie}_2t, \mathbf{ie}_4t, \mathbf{e}_{24}t\}$	$\{t, \mathbf{e}_2t, \mathbf{ie}_4t, \mathbf{ie}_{24}t\}$
4 $\{E_{11} - E_{22} - E_{33} + E_{44} + E_{55} - E_{66} - E_{77} + E_{88},$ $i(-E_{12} - E_{21} + E_{34} + E_{43} - E_{56} - E_{65} + E_{78} + E_{87}),$ $-E_{12} + E_{21} + E_{34} - E_{43} - E_{56} + E_{65} + E_{78} - E_{87},$ $i(-E_{13} - E_{24} - E_{31} - E_{42} + E_{57} + E_{68} + E_{75} + E_{86}),$ $E_{13} + E_{24} - E_{31} - E_{42} + E_{57} + E_{68} - E_{75} - E_{86}\}$	$\{i(-E_{11} + E_{22} + E_{33} - E_{44} - E_{55} + E_{66} + E_{77} - E_{88}),$ $-E_{12} + E_{21} - E_{34} + E_{43} + E_{56} - E_{65} + E_{78} - E_{87},$ $i(E_{12} + E_{21} + E_{34} + E_{43} - E_{56} - E_{65} - E_{78} - E_{87}),$ $i(-E_{13} + E_{24} - E_{31} + E_{42} + E_{57} - E_{68} + E_{75} - E_{86}),$ $E_{13} - E_{24} - E_{31} + E_{42} + E_{57} - E_{68} - E_{75} + E_{86}\}$
5 $\Psi = (s_1 + s_2\mathbf{e}_{45} + \mathbf{is}_3\mathbf{e}_2 + \mathbf{is}_4\mathbf{e}_{245} + \mathbf{is}_5\mathbf{e}_4 + \mathbf{is}_6\mathbf{e}_5$ $+ s_7\mathbf{e}_{24} + s_8\mathbf{e}_{25})t$	$\Psi = (s_1 + s_2\mathbf{e}_{45} + s_3\mathbf{e}_2 + s_4\mathbf{e}_{245} + \mathbf{is}_5\mathbf{e}_4 + \mathbf{is}_6\mathbf{e}_5$ $+ \mathbf{is}_7\mathbf{e}_{24} + \mathbf{is}_8\mathbf{e}_{25})t$
6 $(s_1 + \mathbf{is}_2)E_{11} + (s_3 + \mathbf{is}_4)E_{21} + (s_5 - \mathbf{is}_6)E_{31} + (s_7$ $- \mathbf{is}_8)E_{41} + (s_1 - \mathbf{is}_2)E_{55} + (s_3 - \mathbf{is}_4)E_{65}$ $+ (-s_5 - \mathbf{is}_6)E_{75} + (-s_7 - \mathbf{is}_8)E_{85}$	$(s_1 + \mathbf{is}_2)E_{11} + (s_3 + \mathbf{is}_4)E_{21} + (s_5 - \mathbf{is}_6)E_{31}$ $+ (s_7 - \mathbf{is}_8)E_{41} + (s_1 - \mathbf{is}_2)E_{55} + (-s_3 + \mathbf{is}_4)E_{65}$ $+ (-s_5 - \mathbf{is}_6)E_{75} + (s_7 + \mathbf{is}_8)E_{85}$
7 $\langle \mathbf{e}_1 \widetilde{\Psi}^* \mathbf{e}_1 \Psi \rangle = \frac{1}{4}(s_1 ^2 + \dots + s_8 ^2)$	$\langle \widetilde{\Psi}^* \Psi \rangle = \frac{1}{4}(s_1 ^2 + \dots + s_8 ^2)$

TABLE XV: Complex spinor tables for geometric algebras with vector space dimension $n = 5$ * Electronic address: arturas.acus@tfai.vu.lt† Electronic address: dargys@ftmc.lt¹ P. Lounesto, *Clifford Algebra and Spinors* (Cambridge University Press, Cambridge, 1997), ISBN-13: 978-0521599160.² R. Ablamowicz, *Computer Physics Communications* **115**, 510 (1998).³ R. Ablamowicz and B. Fauser, *Advances in Applied Clifford Algebras* **15**, 157 (2005).

- ⁴ A. Acus and A. Dargys, *Geometric Algebra Mathematica package* (2024), download from <https://github.com/ArturasAcus/GeometricAlgebra>.
- ⁵ D. Hestenes, *Space-Time Algebra* (Gordon and Breach, New York, 1966).
- ⁶ In tables III-IX and X-XV we have used a shorthand notation: $P_1 = \frac{1}{2}(1 + \mathbf{e}_1)$, $P_2 = (-) \equiv \frac{1}{2}(1 - \mathbf{e}_1)$.